What Do Populist Voters Want?
A Nonparametric Revealed Preference Approach

Nina Nikiforova

Bocconi University

November 7, 2026

The New York Times

Monday, September 7, 2026

ANALYSIS

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4 min read

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The party's 44% plurality in Saxony-Anhalt reflects voter frustration with mainstream parties.

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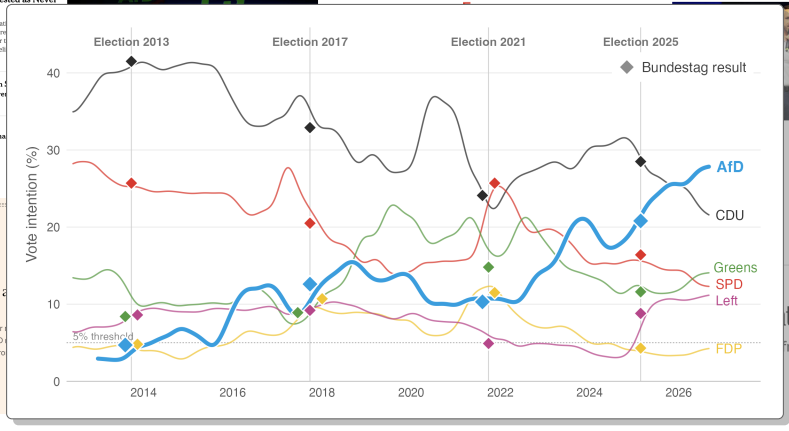
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Motivation

Populists are prevalent – vote share in Western democracies tripled since the 1990s, now $> 1/4$ of the electorate (Guriev and Papaioannou, 2022) – **and harmful for the economy** (Funke et al., 2023)

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Q1. Why? → **What do populist voters want?** Are they aligned with the party on Left–Right (redistribution, immigration, culture) or on Anti-Elite (institutions, direct democracy)?

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What do populist voters want? Are they aligned with the party on Left–Right (redistribution, immigration, culture) or on Anti-Elite (institutions, direct democracy)?

Q2. What now? →

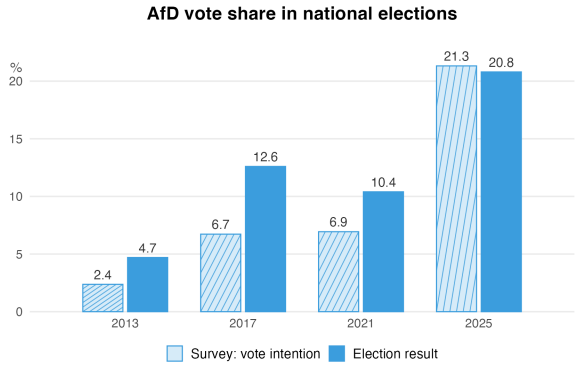
How should establishment parties respond? Can they win populist voters back by moving towards them – and what happens when populists respond?

Recovering preferences of populist voters

1. **Revealed preference**: actual votes, not surveys (misreporting, social desirability)

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1. **Revealed preference**
2. **Multidimensionality**:
   thin (Anti-Elite) and thick (Left–Right) (Mudde, 2004; Guriev and Papaioannou, 2022)

Recovering preferences of populist voters

1. **Revealed preference**
2. **Multidimensionality**
3. **Spatial voting**:
   voters choose the closest party in a weighted policy space (Downs, 1957)

Recovering preferences of populist voters

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Combining 1.–3. is methodologically hard: 1D is well understood, 2D needs strong assumptions

This paper: a nonparametric approach based on partitions of the policy space (Degan and Merlo, 2009; Merlo and De Paula, 2017; Tebaldi et al., 2023)

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Identify and estimate a multidimensional spatial voting model from aggregate vote shares

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- No functional-form assumptions on f: **set identification**

  $\hookrightarrow$ weight w: identified from *which* voting profiles the data require

  $\hookrightarrow$ target parameters (preferences, counterfactual vote shares): sharp bounds via linear programming (Tebaldi et al., 2023)

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- Accommodates selection on observables or instrumental variables

This Paper. Application – Bundestag Elections 2013–2025

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- Party-list vote shares and demographic composition of $\approx$ 11 thous. municipalities per election

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- Original dataset: positions of $\approx$ 4,000 district candidates from two candidate surveys
- Party-list vote shares and demographic composition of $\approx$ 11 thous. municipalities per election
- Results:
  1. AfD support shifts from **anti-elite** to **radical-right**: by 2021, Left–Right explains up to 70% of AfD support; not driven by party repositioning
  2. In 2025, a modest rightward shift by the CDU would attract $\approx 1/5$ of AfD voters – *if the AfD stood still*
  3. Stackelberg game: AfD responds by moving **further right** and reverses the gains $\Rightarrow$ in equilibrium the CDU does not move right

Contribution

Method

1. Structural models of voting from election returns (Kawai and Watanabe, 2013; Ujhelyi et al., 2021; Iaryczower et al., 2024; Kawai and Sunada, 2025; Longuet-Marx, 2026; Berry et al., 2025)
   ⇒ Nonparametric identification & estimation of a multidimensional spatial voting model; relaxes the large-support assumption of Merlo and De Paula (2017) at the cost of set identification
2. Methodologies to retrieve voters' preferences (Fiorina and Abrams, 2008; Draca and Schwarz, 2024; Danieli et al., 2026; Gethin et al., 2022; Desmet et al., 2024)
   ⇒ Revealed preferences from actual election returns instead of surveys

Application

3. Political economy of populism (Algan et al., 2017; Colantone and Stanig, 2018; Margalit, 2019; Autor et al., 2020; Rodrik, 2021; Guriev and Papaioannou, 2022)
   ⇒ Relative importance of ideology vs. anti-elite sentiment, and its evolution
4. Party competition with niche / populist challengers (Palfrey, 1984; Meguid, 2005; Abou-Chadi and Krause, 2020; Spoon and Klüver, 2020; Krause et al., 2023)
   ⇒ First empirical equilibrium analysis of repositioning between an establishment and a populist party

Outline

1. **Model**
   - 1.1. Setup: voters, candidates, elections, election returns
   - 1.2. Identification of ideal-point distributions and counterfactuals
   - 1.3. Identification of the weight
   - 1.4. Scope of the approach
2. **Application**
   - 2.1 Context, data, identification
   - 2.2 Results: weights and distribution of ideal points of AfD voters
3. **Competition: can the CDU win them back?**
   - 3.1 CDU repositioning counterfactuals
   - 3.2 Equilibrium: repositioning game between the CDU and the AfD
4. **Conclusion**

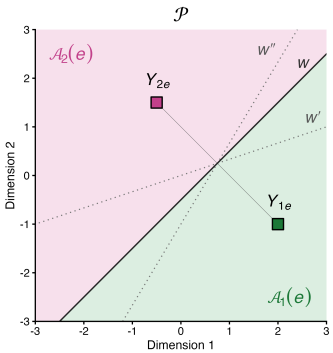
Model

Spatial Voting Model

- Elections $e \in \mathcal{E}$; candidates $j \in \mathcal{J}_e$ with positions $Y_{je} \in \mathcal{P} \subset \mathbb{R}^2$ (known)
- Voter i: ideal point $X_i \in \mathcal{P}$; votes sincerely for the nearest candidate in **weighted** Euclidean distance
  (Downs, 1957; Hinich and Munger, 1996)

$$\text{dist}(X_i, Y_{je}) = \sqrt{(X_i - Y_{je})' W (X_i - Y_{je})}, \quad W = \text{diag}(1, w)$$

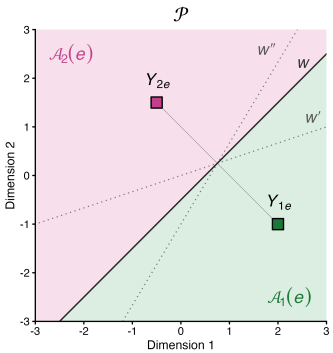
- Choice: $c(X_i, e) = \arg\min_{j \in \mathcal{J}_e} \text{dist}(X_i, Y_{je})$
- **Attraction region** of j in e:
  $\mathcal{A}_j(e) = \{x \in \mathcal{P} : c(x, e) = j\}$



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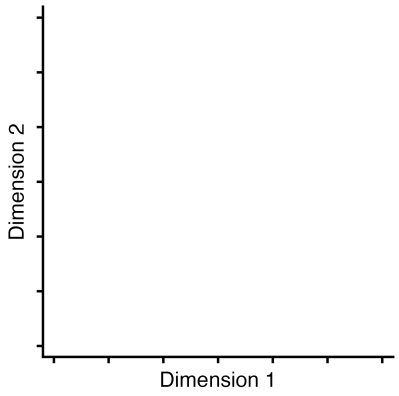
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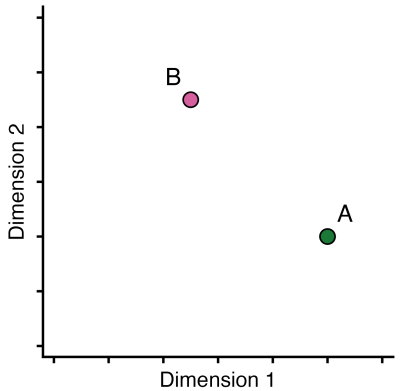
Primitives: the weight w and the distribution of ideal points X_i. What can be learned about them from **aggregate voting data**?

Illustration



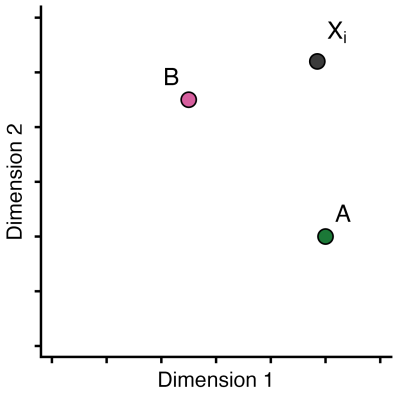
Two-dimensional policy space $\mathcal{P}$

Illustration



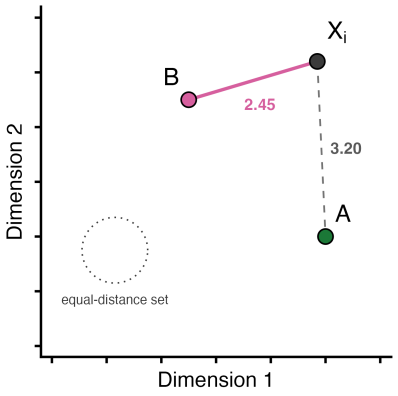
Election e with two candidates, $\mathcal{J}_e = \{A, B\}$, positions $Y_{Ae}, Y_{Be} \in \mathcal{P}$

Illustration



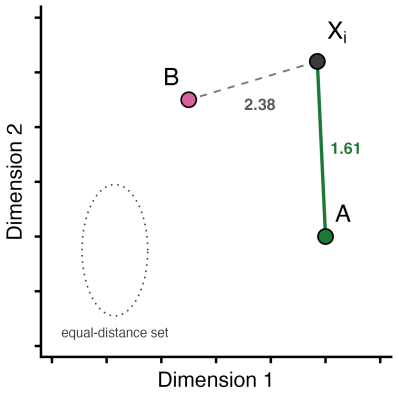
Voter i with ideal point X_i

Illustration



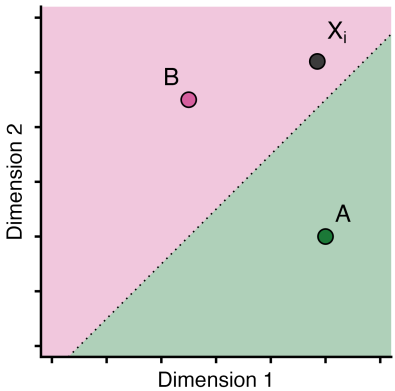
$w = 1$: $\text{dist}(X_i, Y_{Be}) < \text{dist}(X_i, Y_{Ae}) \Rightarrow c(X_i, e) = B$

Illustration



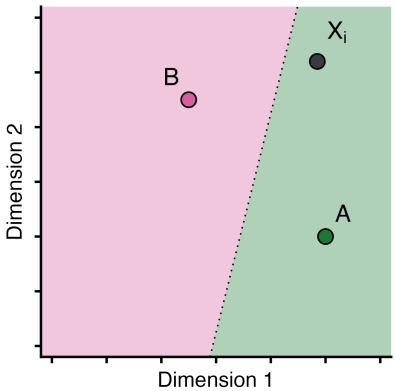
$w = \frac{1}{4}$ (Dimension 2 counts less): $\text{dist}(X_i, Y_{Ae}) < \text{dist}(X_i, Y_{Be}) \Rightarrow c(X_i, e) = A$

Illustration



Same rule at every $x \in \mathcal{P}$: $w = 1$ splits $\mathcal{P}$ at the bisector of $Y_{Ae}Y_{Be}$ into $\mathcal{A}_A(e)$ and $\mathcal{A}_B(e)$

Illustration



A different w rotates the boundary around the midpoint: **different attraction regions**

Election Returns

Econometrician observes vote shares aggregated over administrative units

- Voter i lives in municipality M_i, in state S_i, and has observed socio-demographic type D_i; realizations (s, m, d)
- Each state is divided into electoral districts, each with its own election; municipality m belongs to one district $\Rightarrow$ one election $e(m)$

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$$\mathcal{s}_j(m) \;=\; \sum_d q_{m,d} \int_{\mathcal{A}_j(e(m))} f(x \mid m, d)\, dx$$

Identifying Assumption

Assumption 1a. For every state s, type d, and municipalities m, m' in s:

$$f(x \mid m, d) = f(x \mid m', d) \quad \forall x \qquad \Rightarrow \quad \text{common density } f(x \mid s, d)$$

- Conditional on type, preferences have the same distribution across municipalities within a state
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- **IV version** (Assumption 1b): $f(x \mid z, s, d) = f(x \mid z', s, d)$ for an instrument Z_i that shifts the candidates faced (relevance) but not ideal points (exogeneity); 1a is the case $Z_i = M_i$

Match the Data & Target Parameter

Under Assumption 1a, a density $f \in \mathcal{F}$ predicts the vote shares in m:

$$\sigma_j(m; f) = \sum_d q_{m,d} \int_{\mathcal{A}_j(e(m))} f(x \mid s, d)\, dx$$

Match-the-data (MD): $\underbrace{\hat{s}_j(m)}_{\text{observed}} = \underbrace{\sigma_j(m; f)}_{\text{predicted}} \quad \forall j \in \mathcal{J}_{e(m)},\ \forall m$

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Target parameter $\theta(f)$: functionals of preferences; e.g. share of left-leaning voters, $T = \{x \in \mathcal{P} : x_1 \leq \underline{x}_1\}$:

$$\theta(s, d; f) = \int_T f(x \mid s, d)\, dx$$

Others: share of a candidate's voters with ideal points in a region; vote shares under counterfactual platforms

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Sharp identified sets: $\mathcal{F}^* = \{f \in \mathcal{F} : \text{Assumption 1a and (MD) hold}\}$, $\Theta^* = \{\theta(f) : f \in \mathcal{F}^*\}$

Key Challenge

How to characterize Θ^*?

f is unrestricted $\rightarrow$ $\mathcal{F}$ is infinite-dimensional $\rightarrow$ direct search over $f \in \mathcal{F}$ infeasible

Tebaldi et al. (2023), quasilinear discrete choice: an equivalent **finite-dimensional** problem

$\hookrightarrow$ I adapt their argument to the multidimensional spatial voting model

For now: treat w as known

Minimal Relevant Partition (MRP)

Under Assumption 1a, type d in state s can be treated *as if* a single population voting in every election $e \in \mathcal{E}(s)$

- **Voting profile** $\mathrm{j} = (j_e)_{e \in \mathcal{E}(s)}$: the candidate a voter with a fixed ideal point would choose in each election
- **Cell:** ideal points producing profile j

$$\mathcal{X}_{\mathrm{j}} = \bigcap_{e \in \mathcal{E}(s)} \mathcal{A}_{j_e}(e)$$

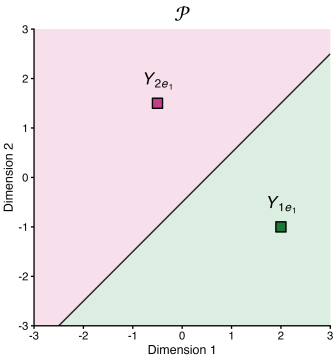
- **MRP:** $\mathbb{X} = \{\mathcal{X}_{\mathrm{j}} : \mathcal{X}_{\mathrm{j}} \neq \varnothing\}$, a partition of $\mathcal{P}$ with one cell per feasible profile

Same cell $\Leftrightarrow$ identical votes in every election; different cells $\Leftrightarrow$ different choice in at least one election

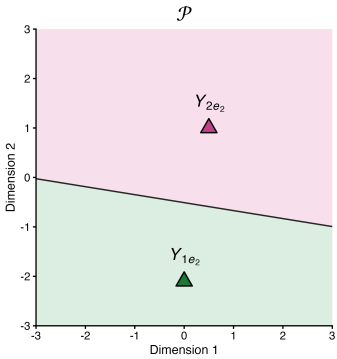
Counterfactual platforms: augment $\mathcal{E}(s)$ by hypothetical elections; regions of $\mathcal{P}$ (e.g. T): refine $\mathbb{X}$ along their boundaries

Illustration

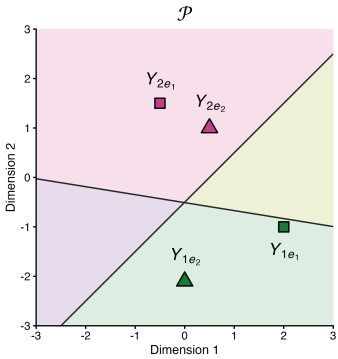
MRP: two elections e_1, e_2 with two candidates each, fixed w



Election e_1: $\mathcal{A}_1(e_1)$, $\mathcal{A}_2(e_1)$



Election e_2: $\mathcal{A}_1(e_2)$, $\mathcal{A}_2(e_2)$



MRP: four cells = profiles
$(1,1), (1,2), (2,1), (2,2)$

From Densities to Cell Masses

Returns depend on f **only through the mass on each cell**:
$\phi(\mathcal{X} \mid s, d) = \int_{\mathcal{X}} f(x \mid s, d)\, dx$

Match-the-data in cell masses (MD′): $\mathbb{X}_j(e) =$ cells contained in $\mathcal{A}_j(e)$

$$\underbrace{\mathcal{Y}_j(m)}_{\text{observed}} = \underbrace{\sum_d q_{m,d} \sum_{\mathcal{X} \in \mathbb{X}_j(e(m))} \phi(\mathcal{X} \mid s(m), d)}_{\text{predicted}} \qquad \forall\, j \in \mathcal{J}_{e(m)},\ \forall m$$

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Result (adapting Prop. 1–2 of Tebaldi et al., 2023): $t \in \Theta^* \Leftrightarrow$ some ϕ satisfies (MD′) and $\bar{\theta}(\phi) = t$. For scalar θ:

$$\Theta^* = \Big[\min_{\phi:\,(\text{MD}')} \bar{\theta}(\phi),\ \max_{\phi:\,(\text{MD}')} \bar{\theta}(\phi) \Big]$$

$\hookrightarrow \bar{\theta}$ linear (or linear-fractional) in $\phi \Rightarrow$ both endpoints by **linear programming**; bounds are **sharp**

Illustration: Target Parameter

Data pin down: masses on the four MRP cells

$$\phi_{(1,1)},\ \phi_{(1,2)},\ \phi_{(2,1)},\ \phi_{(2,2)}$$

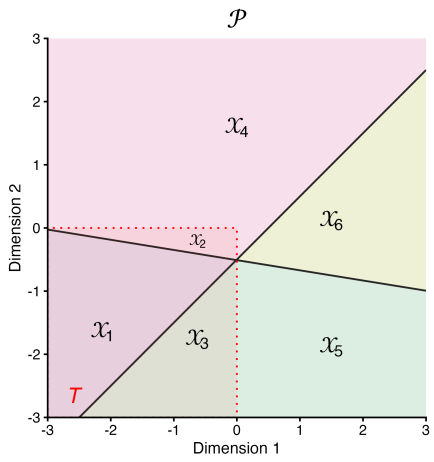


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Target: share of voters in region T (bottom-left quadrant), which refines the MRP into $\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3$

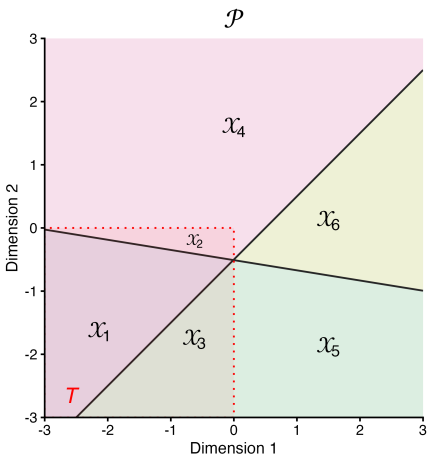


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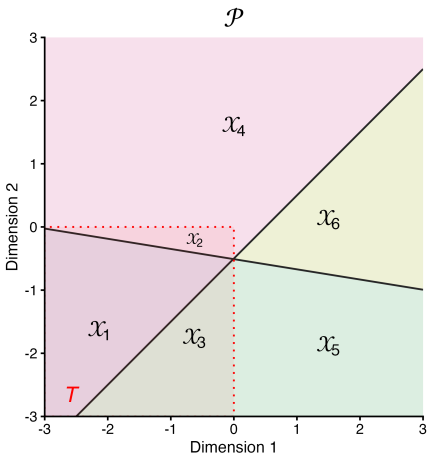
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$$\theta(s, d; f) = \phi(\mathcal{X}_1) + \phi(\mathcal{X}_2) + \phi(\mathcal{X}_3)$$

$$\begin{aligned}
\mathcal{X}_1 = \mathcal{X}_{(2,1)} &\Rightarrow \phi(\mathcal{X}_1) = \phi_{(2,1)} \\
\mathcal{X}_2 \subset \mathcal{X}_{(2,2)} &\Rightarrow \phi(\mathcal{X}_2) \in [0,\ \phi_{(2,2)}] \\
\mathcal{X}_3 \subset \mathcal{X}_{(1,1)} &\Rightarrow \phi(\mathcal{X}_3) \in [0,\ \phi_{(1,1)}]
\end{aligned}$$



Identifying the Weight

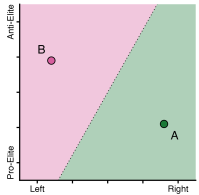
$$\text{dist}(X_i, Y_{je}) = \sqrt{(X_i - Y_{je})' W (X_i - Y_{je})}, \qquad W = \text{diag}(1, w)$$

Identifying the Weight

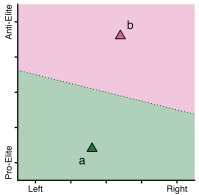
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W

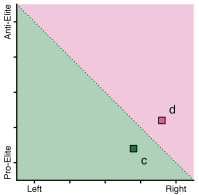
Election e_1



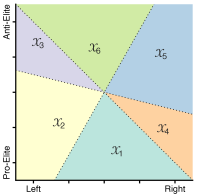
Election e_2



Election e_3

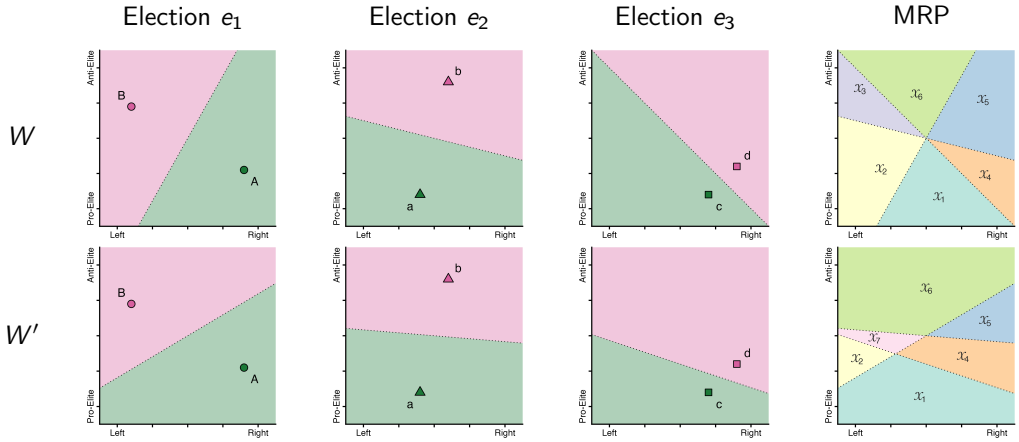


MRP



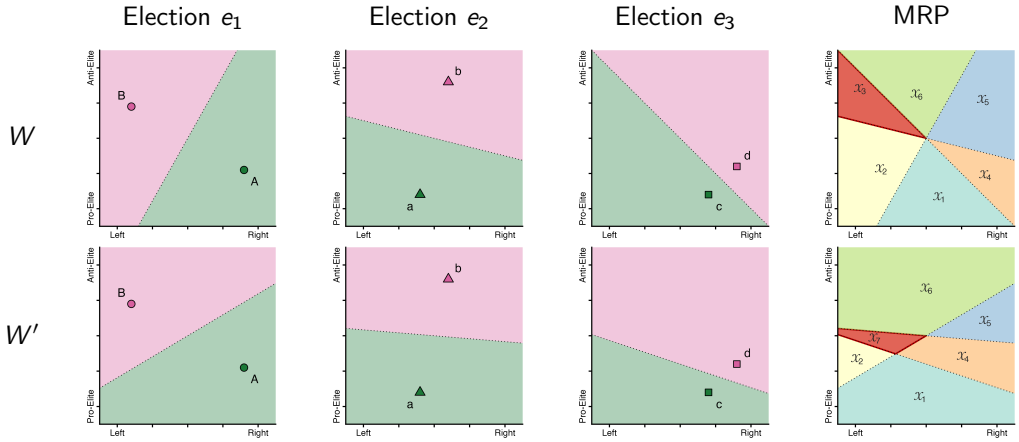
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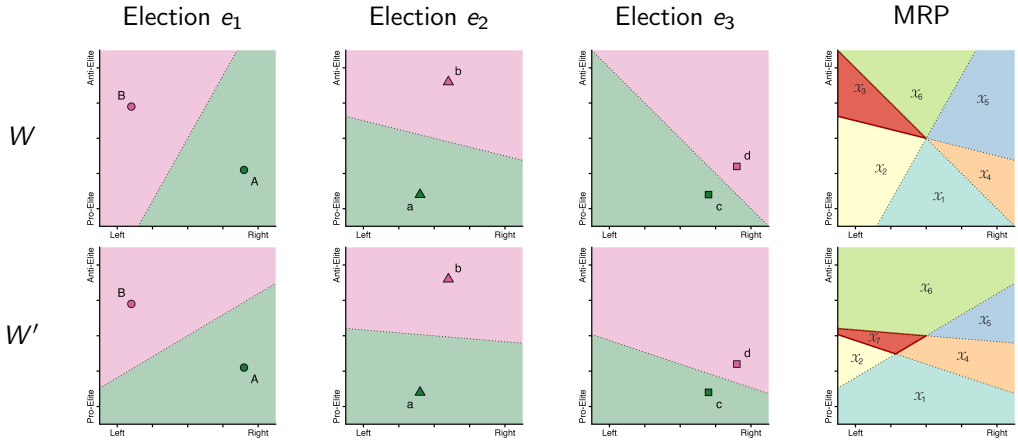
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- The weight defines **which cells** the MRP has
- Identify the weight by the **vote-share fit**

Weights Are Set-Identified

Sharp identified set of weights (adapting Lieber et al., 2024):

$$\mathcal{W}^* = \{w > 0 : \text{some } \phi \text{ on } \mathbb{X}_w \text{ satisfies (MD}')\}$$

- Each profile is available on an open interval of $w \Rightarrow$ feasibility is constant on finitely many intervals
- $\mathcal{W}^*$ is a finite union of open intervals: with finitely many elections, w is **never point-identified**
- Size of $\mathcal{W}^*$ is an empirical question; shrinks with the number of elections simulation

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Two-step characterization of the target parameter:

(1) obtain $\mathcal{W}^*$ (in practice: grid search; fit is flat on the identified set, positive outside)
(2) for each $w \in \mathcal{W}^*$ compute $\Theta^*(w)$ by linear programming

$$\Theta^* = \bigcup_{w \in \mathcal{W}^*} \Theta^*(w), \qquad \text{bounds} \left[\inf_{w \in \mathcal{W}^*} t^*_{lb}(w), \sup_{w \in \mathcal{W}^*} t^*_{ub}(w) \right]$$

Scope of the Approach

Identifying variation: voters with the same preference distribution face different candidates

$f(x \mid z, s, d) = f(x \mid z', s, d)$ for $z \neq z'$ with different candidate positions

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Three sources of such variation in real-world elections:

1. **Simultaneous elections** – one voter, several ballots on the same day (holds by construction)

   US down-ballot races; Australia, Italy (two chambers); Swiss referenda

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   National: Australia, Ireland, Malta. Subnational: Maine, Alaska, several US cities

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   US House, UK Commons; mixed systems: Germany, Japan, Mexico; border designs (Longuet-Marx, 2026)

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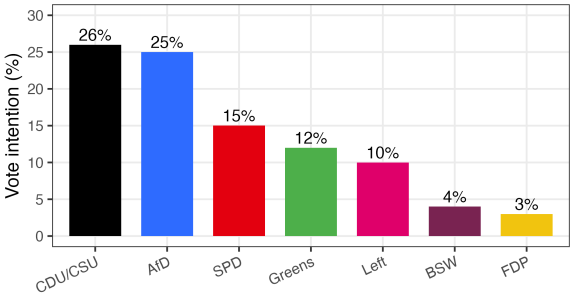
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Extends to: strategic voting (restrict choice sets of viable-candidate voters), valence (additive candidate term)

Application: context & data

Germany's Political Landscape: At a Glance

- AfD tops polls, ending CDU/SPD dominance
- AfD positions itself as radical-right and anti-elite



Note: as of 9 February 2026



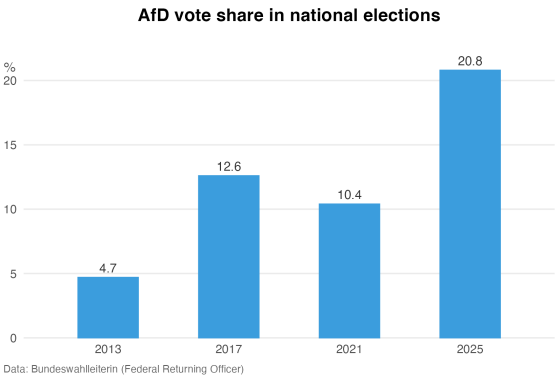
"Our money for our people." **(Right)**



"It's time to eliminate the cronyism of the old parties" **(Anti-Elite)**

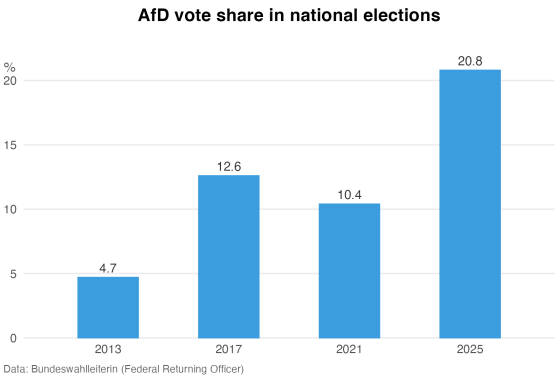
Q. What drives AfD support: ideology or anti-establishment?

The Rise of the AfD



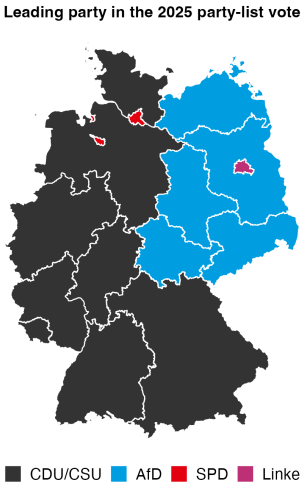
- **Votes:** below the 5% threshold in 2013, second in 2025, first in the polls since

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- **Positions:** the most right-wing *and* most anti-elite party in the sample, in every election (Jolly et al., 2022)

The Rise of the AfD



- **Votes:** below the 5% threshold in 2013, second in 2025, first in the polls since
- **Positions:** the most right-wing *and* most anti-elite party in the sample, in every election (Jolly et al., 2022)
- **East–West divide:** in 2025 the AfD won the most votes in every eastern state; in the West it came second at best

Setting: States, Districts, Municipalities

16 states
preferences estimated within

→

299 districts
own candidates in each

→

≈11,000 municipalities
votes and demographics

Example: Hessen

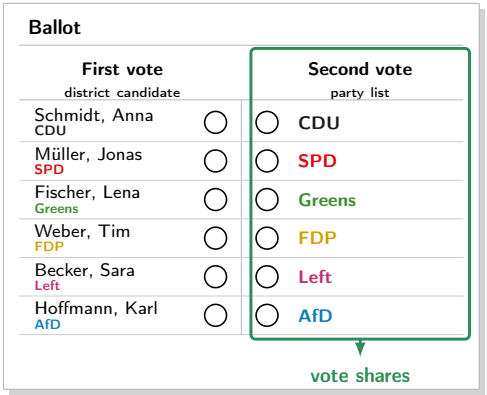


Mixed Electoral System: Two Votes

Ballot

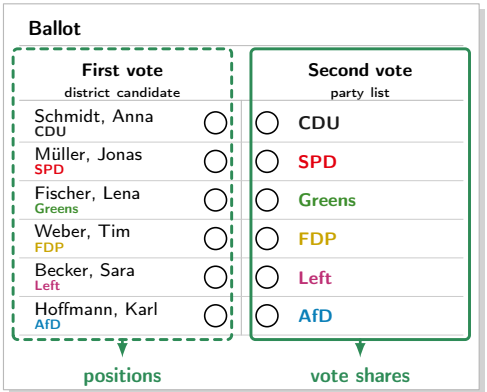
First vote district candidate		**Second vote** party list	
Schmidt, Anna CDU	◯	◯	**CDU**
Müller, Jonas SPD	◯	◯	**SPD**
Fischer, Lena Greens	◯	◯	**Greens**
Weber, Tim FDP	◯	◯	**FDP**
Becker, Sara Left	◯	◯	**Left**
Hoffmann, Karl AfD	◯	◯	**AfD**

Mixed Electoral System: Two Votes



- **Vote shares**: from the party-list column **A1.** List voting is sincere: all six parties $>$ 5% threshold, except FDP and AfD (2013), Left (2021) and FDP (2025) at 4.3–4.9%

Mixed Electoral System: Two Votes



- **Vote shares**: from the party-list column
  A1. List voting is sincere: all six parties $> 5\%$ threshold, except FDP and AfD (2013), Left (2021) and FDP (2025) at 4.3–4.9%

- **Candidate positions**: from the district-candidate column
  A2. The district candidate shapes how the party is perceived locally evidence

Data I: Votes and Voters

Municipal data (municipalities with fewer than 10,000 inhabitants):

1. Party-list vote shares of the six main parties: AfD, CDU/CSU, SPD, Greens, FDP, Left ($> 90\%$ of votes)
2. Joint demographic composition: 24 age–gender–education brackets

   Ex.: *women 18–24 with higher education in Saxony*; *men 60–69 without higher education in Bavaria*

Representative Electoral Statistics: turnout by age $\times$ gender $\times$ state, used to construct the population that turned out example

Data II: Candidate Positions

GLES Candidate Study, every election 2013–2025 ($\approx$ 4,000 district candidates)

- **Left–Right**

  other parties' candidates AfD candidates

  The government should stay out of the economy
  Marriage only for heterosexual couples
  Immigrants should adapt to German culture



- **Anti-Elite**

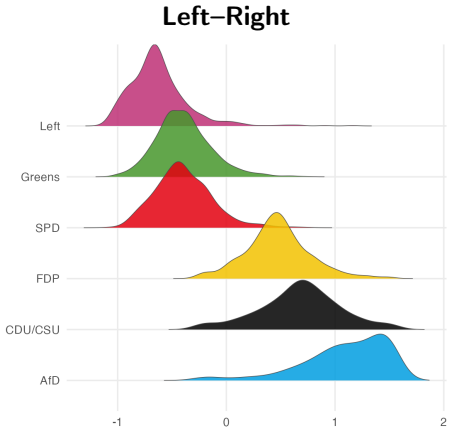
  Dissatisfaction with democracy in Germany
  Legislation does not reflect citizens' interests
  Not enough opportunities to participate

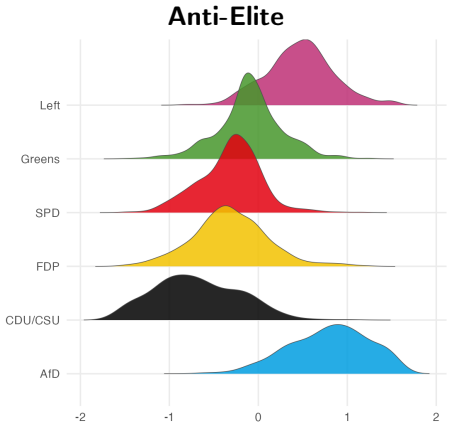


Standardise each item, average within dimension (corr. with IRT > 0.85) all items

Bridge to a second survey, *Wen Wählen?*, by elastic net to improve coverage imputation

Candidate Positions by Party, 2013–2025





medians over time | by year

Identifying Assumption

Within a state s, a demographic group d has the same preferences in every municipality:
$f(x \mid m, d) = f(x \mid m', d)$ for m, m' in state s

Why plausible: state party elites lawfully gatekeep district nominations; about three quarters of district candidates are also on the state lists; low political mobility within states

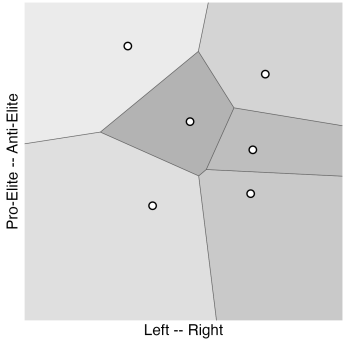
Observationally: within state and party, candidates' positions are uncorrelated with municipal characteristics balance and voters' self-reported views table

Robustness, tested in four ways: results

1. municipalities < 5k; 2. districts where the AfD candidate was not competitive
3. conservative vs. moderate districts allowed to differ; 4. border municipalities only

Example: Partition of the Policy Space

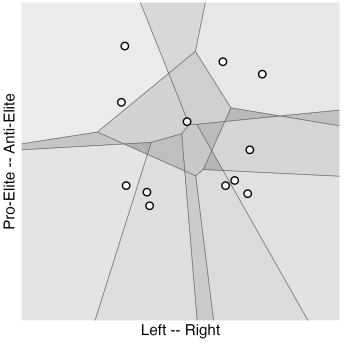
Attraction regions in two electoral districts in Hesse, 2021, and their overlay: estimation



Election e_1 (District 1)



Election e_2 (District 2)

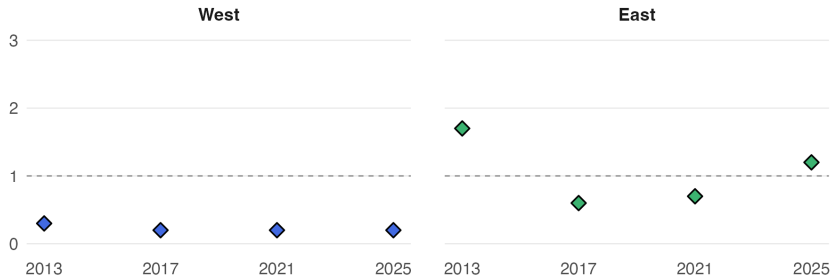


MRP

Results: preferences of AfD voters

How Much Does Anti-Elite Matter Relative to Left–Right?

$$\text{dist}(X_i, Y_{je}) = \sqrt{(X_i - Y_{je})' W (X_i - Y_{je})}, \qquad W = \begin{pmatrix} 1 & 0 \\ 0 & w_{AE} \end{pmatrix}$$

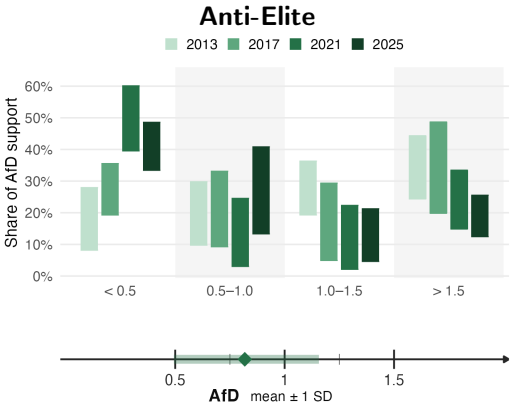


- **West:** Left–Right weighs 3–5 times more than Anti-Elite, in every election
- **East:** Anti-Elite matters throughout; outweighs Left–Right in 2013 and 2025

objective functions confidence sets simulation

Where Are AfD Voters' Ideal Points?

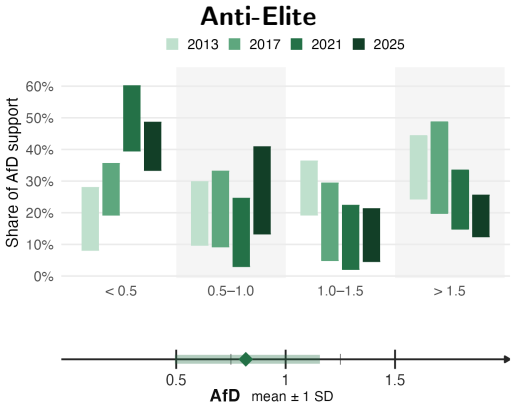
Sharp bounds on the share of AfD voters in each 0.5-wide interval of one dimension

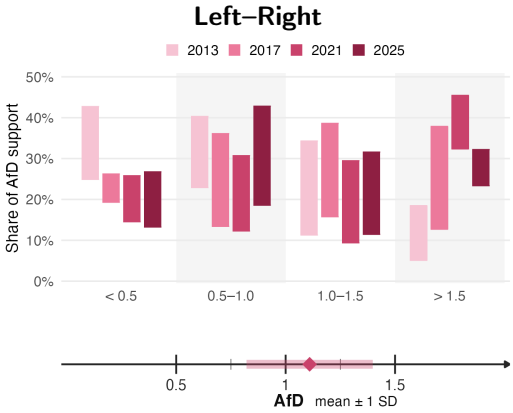


Left–Right

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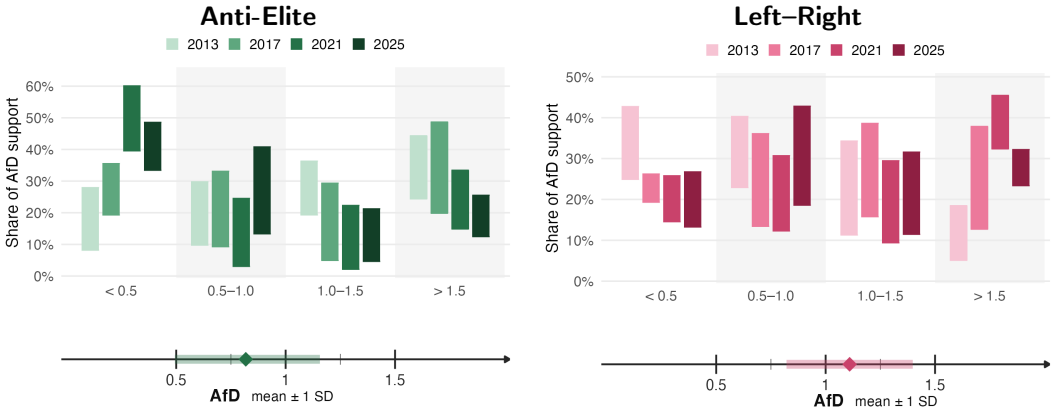
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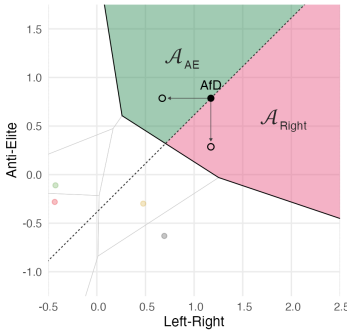


2013: up to 45% strongly anti-elite (> 1.5), at least a quarter only moderately right (< 0.5).
2025: strongly anti-elite share down to about a fifth, radical-right share up sharply. The AfD's own position moved far less.

illustration CIs surveys

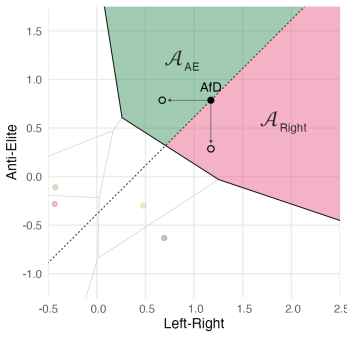
What Motivates the AfD Vote: Protest or Ideology?

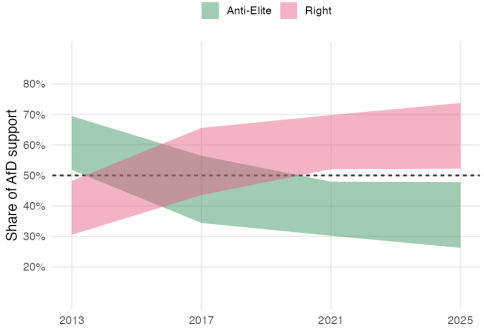
Thought experiment: the AfD splits into a party with the same Left–Right position but less anti-elite, and one as anti-elite but further left. What share of AfD voters prefers each?



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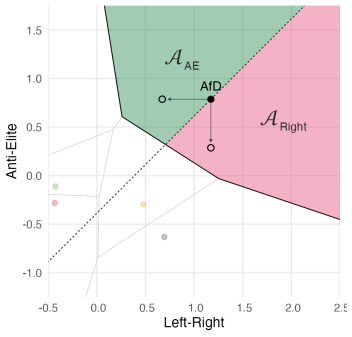
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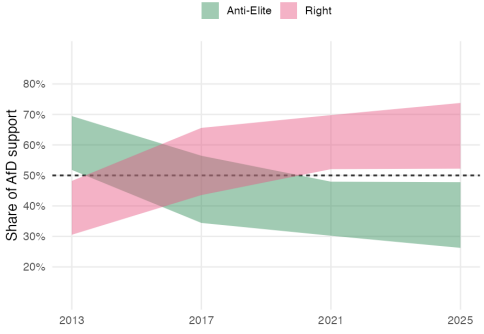




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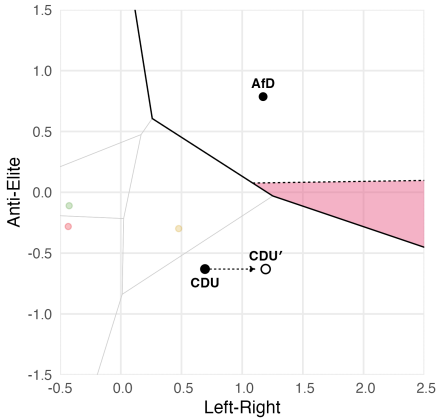


2013: anti-elite majority. 2021–2025: up to 70% radical-right motivated

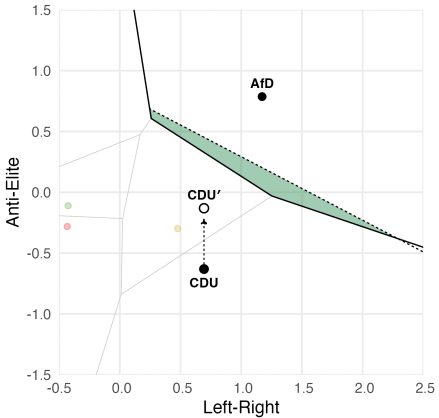
East / West | education, age | robustness | bootstrap CI | weights | vs. surveys | supply or demand? | Merlo–de Paula

Competition: can the CDU win them back?

Can the CDU Win AfD Voters Over?



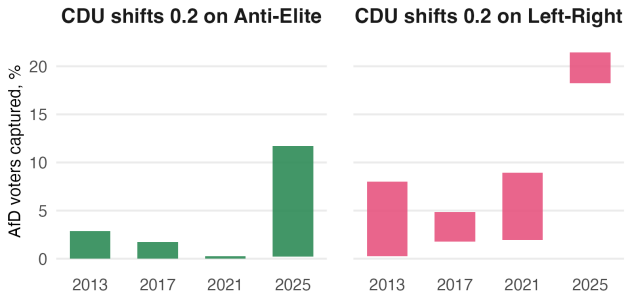
CDU shifts right



CDU shifts anti-elite

Shaded: AfD voters captured by the CDU, holding the AfD and the other parties fixed at their national medians

A Small Shift Right Wins a Fifth of AfD Voters in 2025

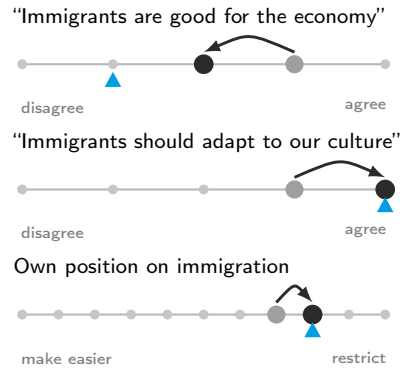


- 0.2 on Left–Right: at most 9% of AfD voters in 2013–2021, **about 20% in 2025**
- Shifting on Anti-Elite is never as effective
- Consistent with the survey experiment of Guenther and Nunnari (2025) and with just over 1m CDU→AfD switchers in 2025 (Schläger et al., 2025)

AfD's own shifts | 0.5 steps | CIs

What Is a 0.2 Shift on Left–Right?

Immigration



● CDU median ▲ AfD median (2025 candidates)

. . . or economic policy | . . . or cultural policy | item by item

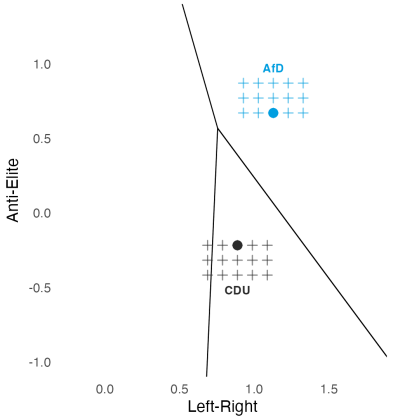
So Why Doesn't the CDU Move?

1. Losses on its **left flank**

So Why Doesn't the CDU Move?

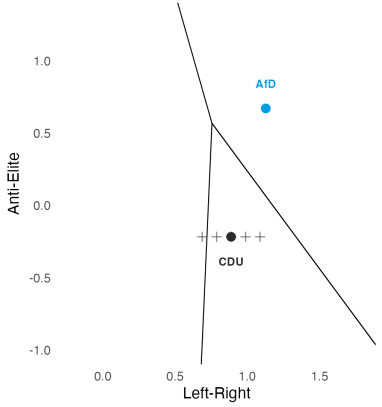
1. Losses on its **left flank**
2. The **strategic response** of the AfD

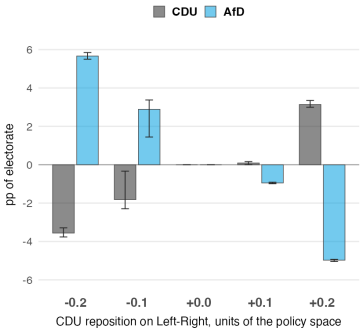
So Why Doesn't the CDU Move? A Repositioning Game, 2025



- **Players:** CDU and AfD; other parties fixed
- **Possible moves**, in steps of 0.1:
  AfD $\{-0.2, \ldots, 0.2\} \times \{0, \ldots, 0.2\}$
  CDU $\{-0.2, \ldots, 0.2\} \times \{-0.2, \ldots, 0\}$
- **Payoff:** change in vote share (a target parameter)
- **Sequential:** the CDU moves first, the AfD responds (Stackelberg; no pure-strategy Nash equilibrium here, cf. Osborne, 2025)
- **Uncertainty:** 100 preference distributions drawn from the identified set; the game is solved for each
- **No turnout margin:** 2025 voters only, abstention held fixed

Scenario 1: The AfD Kept Fixed

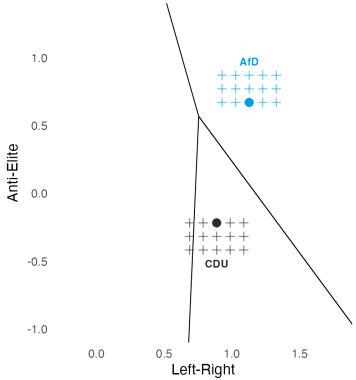




- Moving right pays **net**: the further right, the larger the gain (a 0.1 step $\approx$ 0.3 pp, a 0.2 step $\approx$ 3 pp of the electorate)
- Gains from the AfD exceed losses to the FDP on the left flank

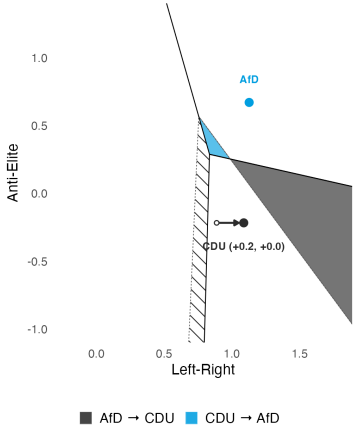
Bars: median net change across 100 draws from the identified set; whiskers: interquartile range. Left–Right moves only.

Scenario 2: The AfD Responds



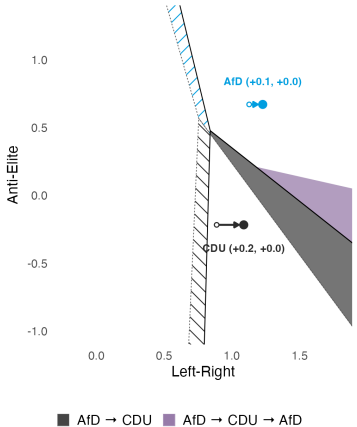
- Both parties can move; the CDU moves first

Scenario 2: The AfD Responds



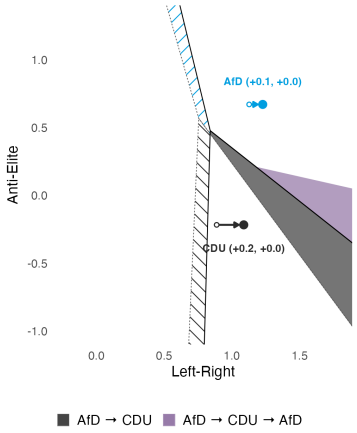
- Both parties can move; the CDU moves first
- Two steps right are net beneficial for the CDU if the AfD does not respond: it captures the pro-elite, far-right wedge

Scenario 2: The AfD Responds



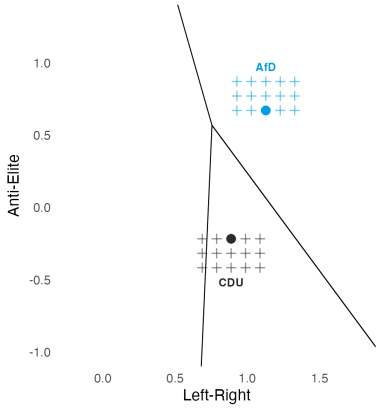
- Both parties can move; the CDU moves first
- Two steps right are net beneficial for the CDU if the AfD does not respond: it captures the pro-elite, far-right wedge
- The AfD **follows one step right**: it wins back the far-right wedge

Scenario 2: The AfD Responds

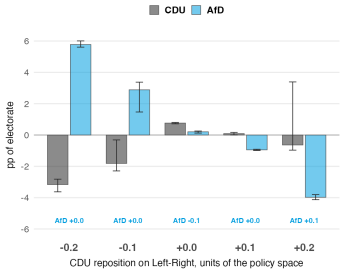


- Both parties can move; the CDU moves first
- Two steps right are net beneficial for the CDU if the AfD does not respond: it captures the pro-elite, far-right wedge
- The AfD **follows one step right**: it wins back the far-right wedge
- The 0.2 move right is **no longer** net profitable for the CDU (-0.6 pp at the median)

Scenario 2: The AfD Responds



CDU net gain after the AfD's best response:



- **Stackelberg equilibrium:** the CDU stays put in 70% of draws (Left–Right moves only) and in 64% when both parties move on both dimensions
  both dimensions
- On both dimensions the AfD replies to CDU $(+0.2, 0)$ with $(+0.1, +0.1)$ or $(+0.1, +0.2)$: harder on Left–Right *and* more anti-elite

Labels under the bars: the AfD's modal reply across draws. Merlo–de Paula firewall

Lessons and Caveats

- **Multidimensionality is essential:** in 1D, a populist's reply to an approaching rival is to move *towards* it (outer flank already its own); in a weighted 2D space, hardening can be a best reply without overtaking
- **What matters is the distribution of ideal points** relative to the boundaries parties can move – recovered here without functional-form assumptions
- Other parties held fixed: competition on each player's outer flank is one-sided; the main answer (once the AfD can respond, moving right is not the CDU's best move) is less sensitive to this
- Anticipating the challenger's reaction: long emphasized in theory (Palfrey, 1984), rarely quantified

Extensions and Robustness

Extensions

- Decomposing Left–Right into **Economics and Culture**: increasing role of culture over economics in AfD support go
- **Coalitions and the firewall:** CDU + AfD is the coalition preferred by a majority, but the welfare ranking is ambiguous go
- **Supply or demand?** The shift towards Right-motivated support is not driven by party repositioning go

Robustness

- More dimensions go
- Positions measured with error go
- Probabilistic voting go
- Method of Merlo and De Paula (2017) go

Conclusion

- **Method**
  - A nonparametric, revealed-preference approach to estimating voters' preferences in a multidimensional spatial voting model
  - Portable to any setting where similar voters face different ballots
- **Preferences**
  - AfD support shifted from anti-elite to radical-right ideology: by 2021, radical-right positions explain up to 70% of AfD support
  - A change in voters, not in the party
- **Establishment response**
  - A small step right on Left–Right (e.g. a harder line on immigration) would win the CDU a fifth of AfD voters in 2025 if the AfD stood still
  - But in equilibrium the AfD responds by moving further to the right and the gain is gone: the CDU is better off not moving right

Thanks!

nina.nikiforova@unibocconi.it

Estimation

Separately by region (East/West) and year; $w = w_{AE}$ common within region-year, $f(x \mid s, d)$ by state and type

Step 1 – weight. For each w_{AE} on a grid $\mathcal{W}$: build the MRP, merge slivers to a fixed mesh of 1,000 cells, solve an LP that best fits the municipal AfD shares; pick the weight with the smallest mean absolute error

$$\hat{w}_{AE} = \arg\min_{w_{AE} \in \mathcal{W}} \left[\min_{\phi \in \Phi_{w_{AE}}} \frac{1}{|\mathcal{M}|} \sum_{m} \left| s_{AfD}(m) - \sigma_{AfD}(m; \phi, w_{AE}) \right| \right]$$

Step 2 – bounds. Fix $\hat{w}_{AE}$; best attainable fit $\hat{Q}$; bounds on θ = min / max over masses whose error is within $\epsilon_{abs} = 10^{-5}$ of $\hat{Q}$ (Mogstad et al., 2018; Tebaldi et al., 2023)

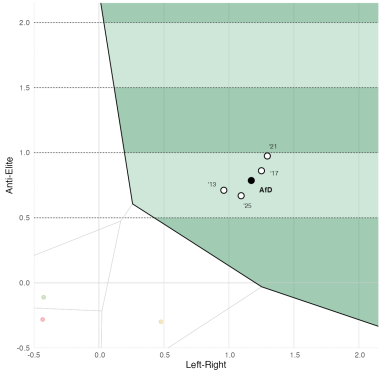
details

Model fit: correlation of predicted and observed municipal AfD shares > 0.9 in 2017–2025, 0.59 in 2013; reproduces how AfD support varies with local demographics

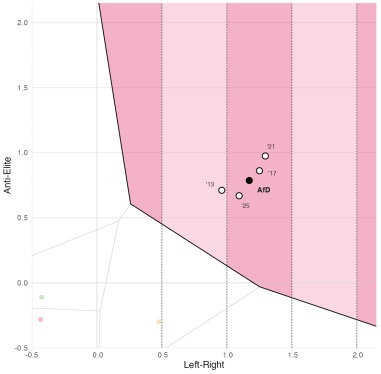
back

fit

Intervals of One Dimension in the Policy Space



Anti-Elite intervals

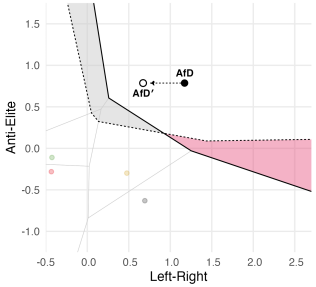


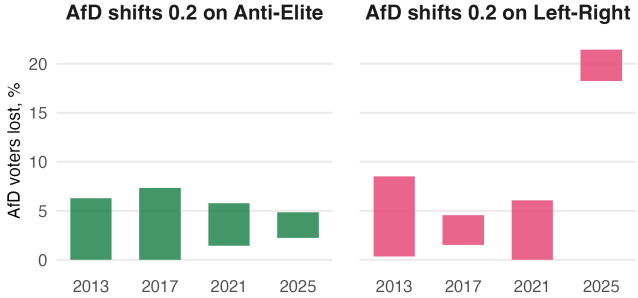
Left–Right intervals

Each interval is a band across the AfD attraction region; adjacent bands share mesh cells, so shares can sum to more than one within a year.

back

How Sensitive Is AfD Support to the AfD's Own Moderation?



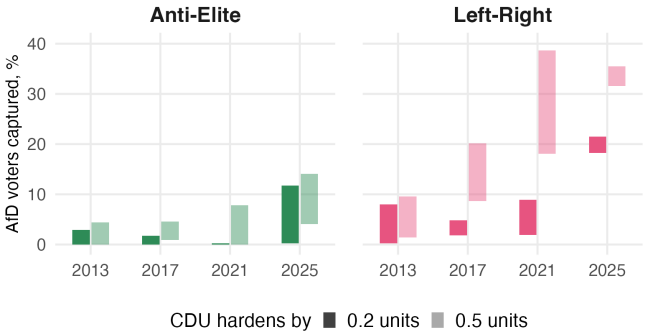


voters lost (pink), 0.5 shift

- Share of baseline AfD voters lost when the AfD moderates by 0.2 on one dimension, other parties fixed illustration 0.5 steps, CIs
- **Before 2025:** a 0.2 step costs $\leq 9\%$ of voters (0.5: $\leq 19\%$). **2025, Left–Right:** 18–22% (0.5: 35–48%); **Anti-Elite:** $\leq 5\%$

back

CDU Shifts of 0.2 and 0.5



Share of baseline AfD voters captured by the CDU; larger rightward steps have become increasingly effective over time, anti-elite steps never as effective.

back

Supply or Demand? Shapley Value of Party Positions

Change in the Right share θ_{Right} between consecutive elections decomposed into preferences (Pref), composition (Comp; Representative Electoral Statistics) and party positions (Pos):

$$\theta_{Right}(Pref_{t'}, Comp_{t'}, Pos_{t'}) - \theta_{Right}(Pref_t, Comp_t, Pos_t) = \phi(Pref) + \phi(Comp) + \phi(Pos)$$

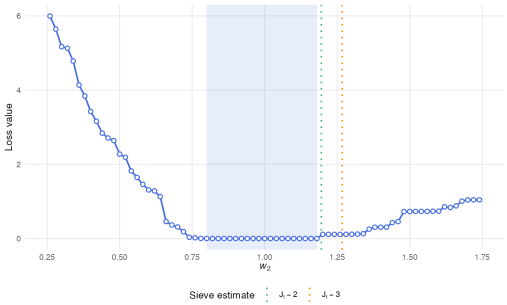
Bounds on $\phi(Pos)$	2013→2017	2017→2021	2021→2025
West	$[-0.202, -0.026]$	$[-0.047, +0.000]$	$[+0.023, +0.117]$
East	$[-0.103, -0.018]$	$[+0.000, +0.018]$	$[-0.041, -0.005]$
National	$\mathbf{[-0.188, -0.025]}$	$\mathbf{[-0.038, +0.004]}$	$\mathbf{[+0.008, +0.092]}$

- Positions contribute positively only in 2021–2025 ⇒ **the shift towards Right-motivated support is not primarily driven by party repositioning**
- Bounds on the Pref and Comp channels are wide and uninformative details back

Weights: Simulation

10,000 voters, 24 six-candidate elections, true $w = 1$; ideal points from a two-cluster mixture; only vote shares observed

- Fit is exactly zero on $[0.80, 1.18]$ (contains the truth), positive outside: **cell geometry alone set-identifies** w
- Sieve estimates of Merlo and De Paula (2017) ($J_t = 2, 3$): 1.20, 1.27 – outside the identified set
- Identified set shrinks with the number of elections: $[0.58, 1.18]$ with 10, $[0.93, 1.05]$ with 100
- Noisy perceptions of positions ($\sigma = 0.1$): strict interior minimum; near-feasible set $[0.98, 1.06]$; sieve 1.09, 1.40
- Removing up to 50% of the smallest cells leaves the identified set essentially unchanged



back

Estimation – Step 1: Weight

Notation. Region × year; states $s \in \mathcal{S}$, types $d \in \mathcal{D}$, municipalities $m \in \mathcal{M}$; mesh $\{\mathcal{X}_1, \ldots, \mathcal{X}_n\}$, $n = 1{,}000$, rebuilt at every w_{AE}
$\phi_{s,d} \in \mathbb{R}^n_+$, $1^\top \phi_{s,d} = 1$: cell masses of type d in state s; $a_{m,\ell}(w) \in \{0,1\}$: cell ℓ in the AfD attraction region in $e(m)$

Predicted AfD share in m:

$$\sigma_{AfD}(m; \phi, w) = \sum_{d \in \mathcal{D}} q_{m,d}\, \boldsymbol{a}_m(w)^\top \phi_{s(m),d}$$

Best-fit LP with slacks $u_m \geq 0$ (Mogstad et al., 2018; Tebaldi et al., 2023):

$$\bar{Q}(w_{AE}) = \frac{1}{|\mathcal{M}|} \min_{\phi, \boldsymbol{u}} 1^\top \boldsymbol{u} \quad \text{s.t.} \quad -u_m \leq \sigma_{AfD}(m; \phi, w) - \text{𝒹}_{AfD}(m) \leq u_m \;\; \forall m$$

$$\hat{w}_{AE} = \arg\min_{w_{AE} \in \mathcal{W}} \bar{Q}(w_{AE})$$

Grid $\mathcal{W} = \{0.2, \ldots, 1\} \cup \{1/0.9, \ldots, 1/0.2\}$. In the population $\bar{Q} = 0$ on the identified set; with noise a unique minimizer, reported as a representative value. Merging slivers keeps noise in positions from propagating, fixes the number of cells across weights, and keeps the LP tractable. Full-partition results are close full MRP.

Estimation – Step 2: Bounds on Target Parameters

Fix $\hat{w}_{AE}$. Reference points $C^{AE} = (\bar{Y}_{LR} - 1, \bar{Y}_{AE})$, $C^{Right} = (\bar{Y}_{LR}, \bar{Y}_{AE} - 1)$ split the AfD region; $\mathcal{C}^{AE}, \mathcal{C}^{Right}$ = cells in each area

Mass of AfD voters in cells $\mathcal{C}$:

$$W(\phi; \mathcal{C}) = \sum_s \sum_d \boldsymbol{b}_{s,d}(\mathcal{C})^\top \phi_{s,d}, \qquad b_{s,d,\ell}(\mathcal{C}) = 1\{\ell \in \mathcal{C}\} \sum_{m \in \mathcal{M}(s)} N_m \, q_{m,d} \, a_{m,\ell}$$

Target (linear-fractional): $\theta_{Right}(\phi) = \dfrac{W(\phi; \mathcal{C}^{Right})}{W(\phi; \mathcal{C}^{Right}) + W(\phi; \mathcal{C}^{AE})}$

Coupling across states (loose): on six coarse marginals (education, young/old, gender) and 0.5-blocks B, each state's share within a factor $1 + \tau_{state}$ ($\tau_{state} = 2$) of the regional mean

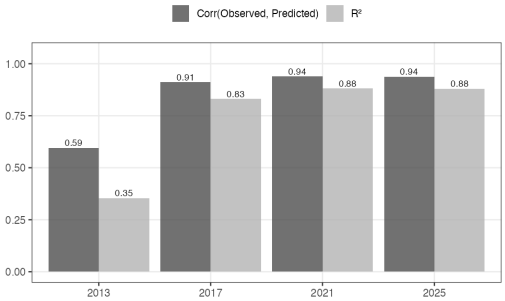
Bounds: best fit $\hat{Q}$; then

$$\underline{\theta}_{Right} = \min \theta_{Right}(\phi), \quad \overline{\theta}_{Right} = \max \theta_{Right}(\phi) \quad \text{s.t. fit} \le \hat{Q} + \epsilon_{abs}, \;\; \epsilon_{abs} = 10^{-5}$$

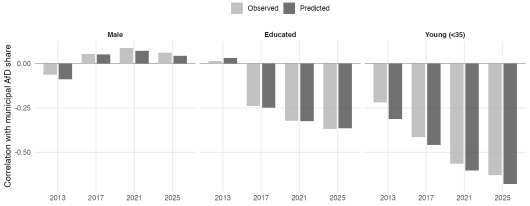
solved by bisection on t: feasibility of the LP with $W(\cdot; \mathcal{C}^{Right}) \le t\,[W(\cdot; \mathcal{C}^{Right}) + W(\cdot; \mathcal{C}^{AE})]$

back

Model Fit



Correlation and R^2, predicted vs. observed AfD shares



Correlation of predicted / observed shares with municipal demographics

Residual s.d.: 0.016 (2013), 0.033 (2017), 0.032 (2021), 0.048 (2025). Holding out 10% of municipalities: merging education or age groups raises the held-out error ($p < 0.01$); gender less so ($p = 0.04$).

back

Data on Candidate Positions I.

Data Source 1 – GLES Candidate Study 2013, 2017, 2021, 2025

- Candidates of the six parties; same political-attitude items as the voter survey (GLES)
- Left–Right: redistribution, state intervention, immigration, abortion, same-sex marriage, self-placement; Anti-Elite: satisfaction with democracy, trust, referendums (up to 7 items, 5 in 2025)
- Each item z-scored with its pooled mean/sd across waves; candidate score = mean of available standardized items (corr. with IRT > 0.85)
- Response rate among district candidates $\approx 35\%$

External validation: within-party evolution broadly consistent with the Chapel Hill Expert Survey (Bakker et al., 2015) (CDU's rightward shift 2021–2025; AfD's hardening 2017–2021)

back

Data on Candidate Positions II.

Data Source 2 – WEN WÄHLEN? (*Whom To Vote For?*): nonpartisan platform with a standardized questionnaire (50–76 items per wave)

- Merge at the district–party level; elastic net per wave and dimension, hyperparameters by 10-fold CV
- Out-of-fold fit: $R^2 = 0.76$ (LR), 0.53 (AE); RMSE/s.d. 0.49 and 0.69
- 1,147 training candidates, 1,348 imputed; combined coverage $\approx 60\%$ of district$\times$party cells (CDU 37%, Greens 77%, AfD $\approx 50\%$)
- Respondents and non-respondents obtain comparable first-vote shares within party
- Remaining positions: within-state party median (national median $< 2\%$); keep districts with ≤ 3 imputed candidates $\Rightarrow \approx 2/3$ of districts

back

Classification questions

Dimension	Question	Values
Anti-Elite	Satisfaction with democracy in Germany	1–4
Anti-Elite	Democracy offers enough opportunities for citizens to participate	1–5
Anti-Elite	Laws reflect citizens' interests	1–5
Anti-Elite	Political parties are the central intermediary between citizens and government	1–5
Anti-Elite	There should be nationwide referendums at the federal level	1–5
Anti-Elite	Referendums tend to lead to bad laws	1–5
Anti-Elite	Citizens have lost trust in politicians	1–5
Left–Right	In a crisis, the government should spend to stimulate economic growth	1–5
Left–Right	In a crisis, Germany should provide financial support to other countries/EU partners	1–5
Left–Right	Immigrants should adapt to German culture	1–5
Left–Right	Government should stay out of the economy	1–5
Left–Right	Same-sex marriage should be banned	1–5
Left–Right	Providing social security should be a key goal of government	1–5
Left–Right	Government should reduce income differences	1–5
Left–Right	Immigration is good for the German economy	1–5
Left–Right	Women should be free to decide about abortion	1–5
Left–Right	Self-placement on the left–right scale	1–11
Left–Right	Immigration policy: make immigration easier vs. restrict it	1–11

back

Item Potency (2017)

Survey item	Theme	Scale	σ (1 pt)
Anti-Elite			
Satisfaction with how democracy works	Current instit.	1–4	0.175
Parties are citizens' central intermediaries	Current instit.	1–5	0.144
Legislation reflects citizens' interests	Current instit.	1–5	0.137
Referendums lead to badly designed laws	Direct democracy	1–5	0.138
Left–Right			
Immigrants are good for the German economy	Immigration	1–5	0.093
Same-sex marriage should be banned	Culture	1–5	0.090
Women should decide themselves about abortion	Culture	1–5	0.085
Providing social security is a core government task	Econ. policy	1–5	0.081
Left–right self-placement		1–11	0.038

σ: change in the standardized dimension score from moving one item by one scale point; a 0.2 (0.5) step requires about $0.2/\sigma$ ($0.5/\sigma$) points on a single item.

back

What Is a 0.2 Shift on Left–Right? Economic Policy

Economic policy

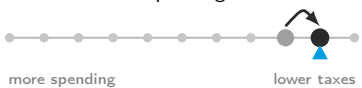
"Government should reduce income differences"



"Government should stay out of the economy"



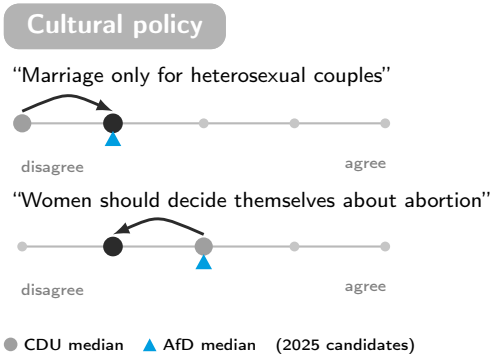
Taxes vs. social spending



● CDU median ▲ AfD median (2025 candidates)

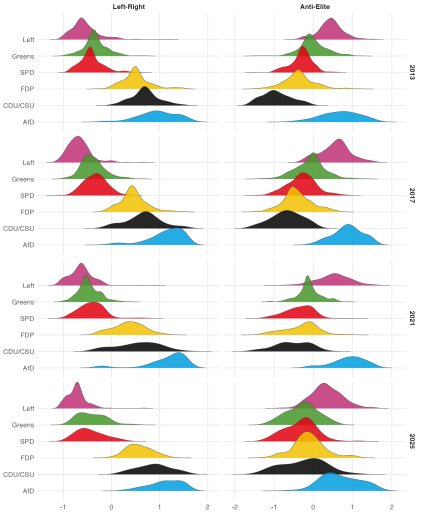
back

What Is a 0.2 Shift on Left–Right? Cultural Policy



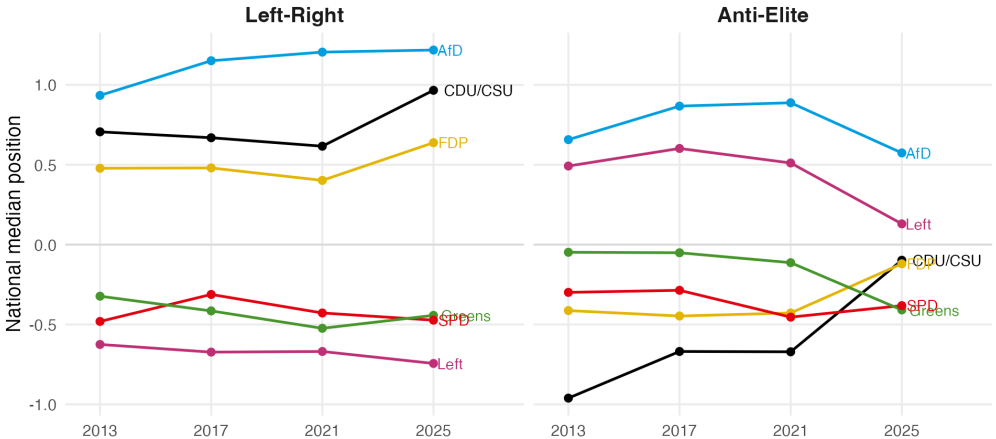
back

Candidate Positions by Party and Election Year



back

Party Medians over Time



back

Candidate positions as proxies of parties' positions

Rationale for the assumption:

(i) **Candidates' positions in electoral districts predict how voters in those districts perceive party positions** (next slide); no effect on voters' own self-placement, which speaks against local sorting

(ii) Voters know their local candidates, report being contacted by them (GLES 2021)
figure

(iii) Candidates and lists overlap: $\approx 2/3$ of list candidates also run in a district; district candidates rank higher on lists

(iv) "Contamination effect" – candidates' characteristics affect parties' list-tier performance in mixed systems (Ferrara et al., 2005)

back

Candidate positions as proxies of parties' positions

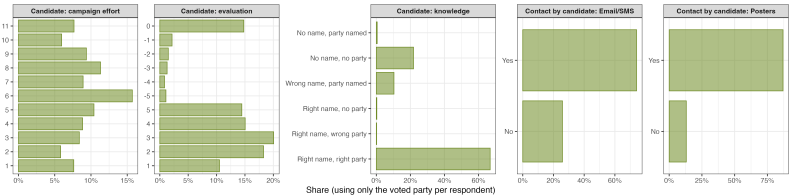
	Perceived Party Position, L-R			*Own L-R Position*
	All Parties	Chosen Parties	Chosen AfD	Chosen Parties
	(1)	(2)	(3)	(4)
Panel A: no individual controls				
Candidate, L-R	−0.007	0.276**	1.207***	0.106
	(0.080)	(0.135)	(0.213)	(0.166)
Panel B: with demographic controls				
Candidate, L-R	−0.015	0.359**	1.243**	0.174
	(0.078)	(0.140)	(0.408)	(0.169)
State F.E.	Yes	Yes	Yes	Yes
Party F.E.	Yes	Yes	No	Yes
Observations	19,992	2,269	210	2,219

GLES 2021 pre-/post-election cross-sections, survey weights, estimation-sample districts. Dependent variable (1)–(3): respondent's placement of the party on the 1–11 scale; (4): own self-placement. SE clustered by party × state. *p<0.1; **p<0.05; ***p<0.01.

back

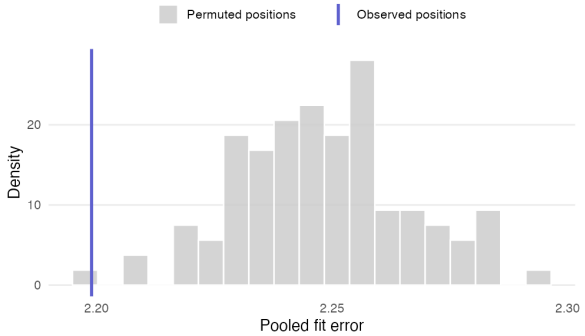
Voters know & interact with local candidates

- **Campaign effort (1–11):** 1 = mostly about the candidate; 11 = mostly about the party
- **Candidate knowledge:** ability to name the constituency candidate and their party
- **Contact:** whether the respondent was contacted via the listed means



back

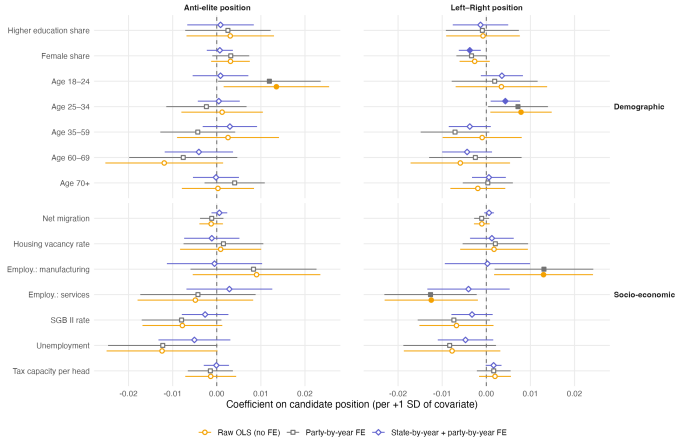
Permutation Test on Fit



Candidate positions randomly reassigned across districts within each state, 100 permutations per region-year; pooled fit error = geometric mean of the eight region-year errors. The observed assignment ranks second of 101 ($p = 0.02$).

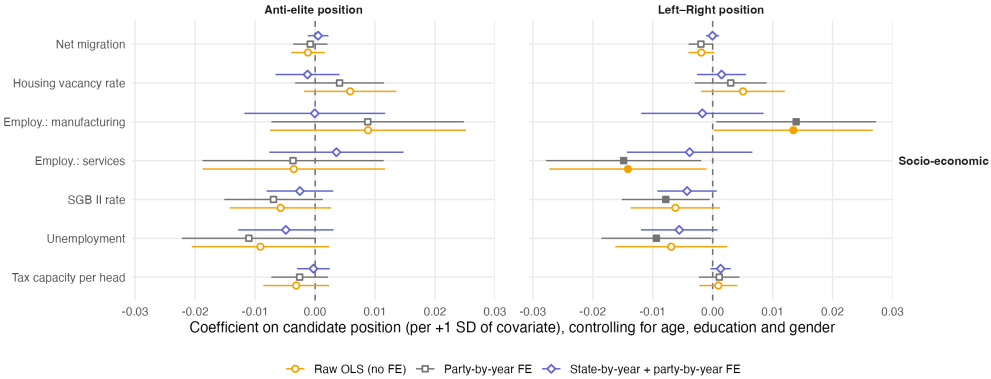
back

Cand. Positions and Municipalities' Characteristics



Each marker: coefficient from a regression of a candidate position on one standardized characteristic; municipality × party cells, 2013–2025. Specifications: raw; party × year FE; state × year + party × [back] FE.

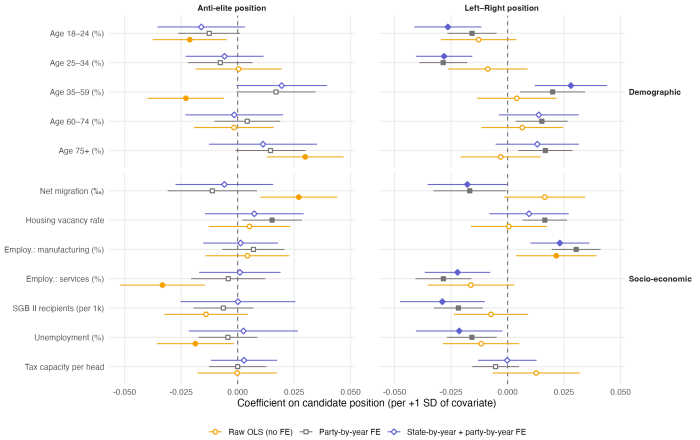
Cand. Positions and Municipalities' Characteristics (II)



Same as previous slide, controlling for the municipality's share with higher education, female share and age-band shares.

back

Cand. Positions and Districts' Characteristics



At the district level positions do correlate with socio-economic characteristics with the expected signs; most associations are muted conditional on demographics.

back

Cand. Positions and Voters' Reported LR Views

	Candidate LR Position		
	All Parties	Chosen Parties	Chosen AfD
	(1)	(2)	(3)
Panel A: no individual controls			
Self, LR	0.0008	0.0023	0.0005
	(0.0011)	(0.0037)	(0.0171)
Panel B: with demographic controls			
Self, LR	0.0017	0.0039	−0.0000
	(0.0012)	(0.0040)	(0.0169)
State F.E.	Yes	Yes	Yes
Party F.E.	Yes	Yes	No
Observations	19,450	2,219	205

GLES 2021, survey weights; self-reported views from 1 (very left) to 11 (very right); SE clustered by party × state.

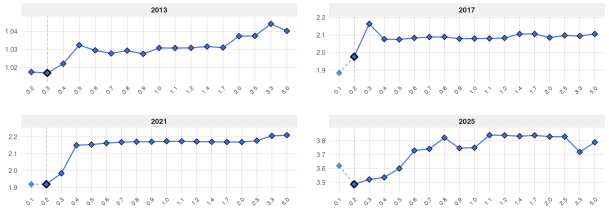
back

Representative Electoral Statistics

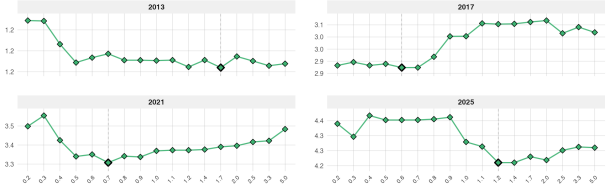
Partei	Insgesamt		davon Wählerinnen und Wähler in der Geburtsjahresgruppe											
			1997 – 2003		1987 – 1996		1977 – 1986		1962 – 1976		1952 – 1961		1951 und früher	
			entspricht Altersgruppe etwa von ... bis ... Jahren											
			18 – 24		25 – 34		35 – 44		45 – 59		60 – 69		70 und mehr	
	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%
	Mecklenburg-Vorpommern													
	Insgesamt													
Insgesamt	**935,0**	***100***	**52,1**	***100***	**91,9**	***100***	**135,3**	***100***	**241,1**	***100***	**205,3**	***100***	**209,4**	***100***
ungültig	**17,0**	***1,8***	0,9	*1,7*	0,9	*1,0*	2,2	*1,6*	3,2	*1,3*	4,1	*2,0*	5,8	*2,8*
gültig	**918,1**	***98,2***	51,2	*98,3*	91,0	*99,0*	133,1	*98,4*	237,9	*98,7*	201,2	*98,0*	203,7	*97,2*
CDU	**180,7**	***19,7***	5,3	*10,4*	11,2	*12,3*	22,2	*16,7*	48,9	*20,6*	42,9	*21,3*	50,0	*24,6*
SPD	**261,4**	***28,5***	9,7	*18,9*	18,8	*20,7*	29,0	*21,8*	59,6	*25,1*	64,4	*32,0*	80,0	*39,3*
AfD	**170,0**	***18,5***	7,4	*14,4*	19,2	*21,1*	33,5	*25,2*	52,7	*22,1*	35,7	*17,7*	21,5	*10,6*
FDP	**66,3**	***7,2***	8,4	*16,4*	9,1	*10,0*	11,5	*8,6*	18,8	*7,9*	11,5	*5,7*	7,0	*3,4*
DIE LINKE	**116,0**	***12,6***	6,7	*13,1*	9,7	*10,7*	12,5	*9,4*	25,3	*10,7*	28,3	*14,1*	33,5	*16,4*
GRÜNE	**62,7**	***6,8***	8,4	*16,5*	12,7	*14,0*	11,4	*8,5*	15,0	*6,3*	8,9	*4,4*	6,3	*3,1*
CSU	**–**	**–**	–	–	–	–	–	–	–	–	–	–	–	–
Sonstige darunter:	**61,0**	***6,6***	5,3	*10,3*	10,3	*11,3*	13,1	*9,9*	17,5	*7,3*	9,5	*4,7*	5,4	*2,6*
FREIE WÄHLER[1]	**18,5**	***2,0***	0,9	*1,7*	2,9	*3,2*	3,2	*2,4*	5,5	*2,3*	4,1	*2,0*	1,9	*0,9*
dieBasis[1]	**15,4**	***1,7***	0,8	*1,5*	1,9	*2,1*	3,8	*2,8*	5,3	*2,2*	2,2	*1,1*	1,4	*0,7*

back

Objective Function at Different Weights



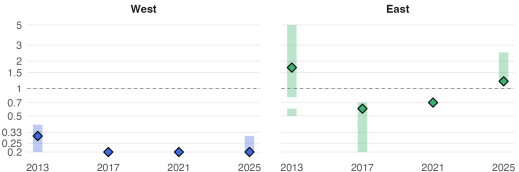
West



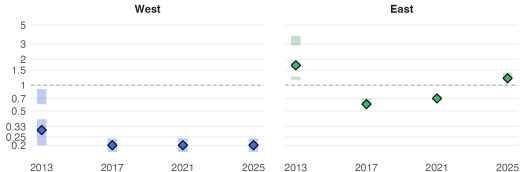
East

back

Weights: Identified Sets and Confidence Sets



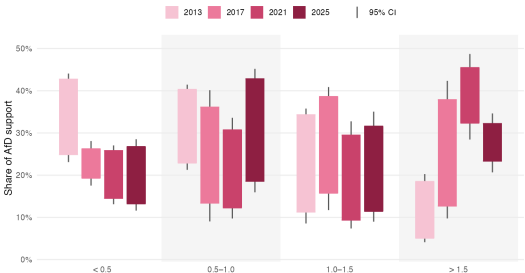
Weights within 1% of the minimum fit error



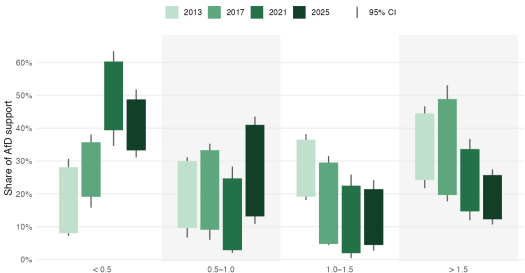
Confidence sets calibrated from $B = 10$ subsample re-runs ($\alpha = 0.05$)

back

Dimension Bins: Sampling Uncertainty



Left–Right

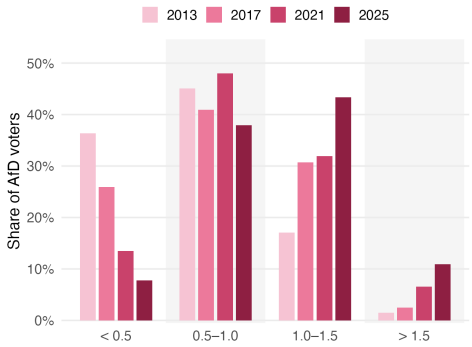


Anti-Elite

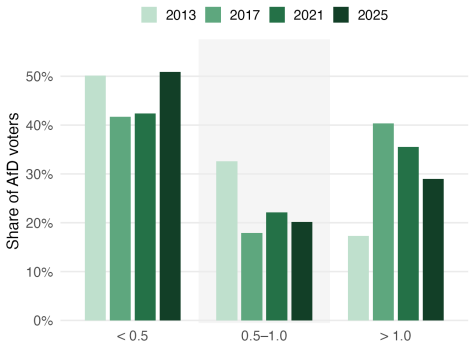
Bars: bounds; gray lines: 95% Imbens and Manski (2004) CIs from a municipality bootstrap ($B = 50$) within districts.

back

Stated Preferences: AfD Voters in the GLES Voter Surveys



Left–Right

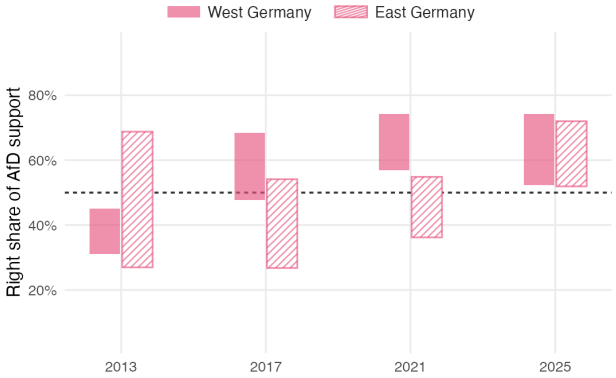


Anti-Elite

Self-reported positions of AfD voters (weighted). Qualitatively consistent with the revealed-preference estimates: shift to the right, decline in anti-elite attitudes over 2017–2025; survey places fewer voters at extreme positions. Anti-Elite measured with different items than for candidates.

back

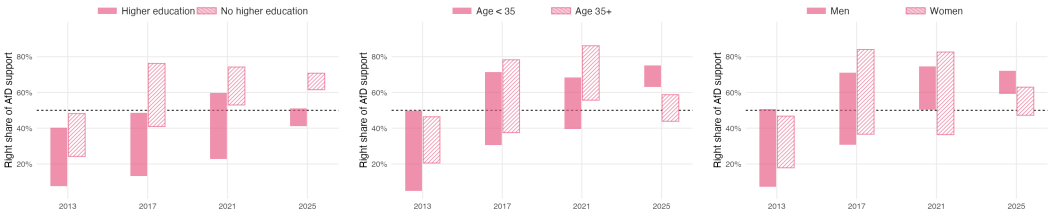
Motivations: East/West



Bounds on θ_{Right} separately for West (solid) and East (hatched)

back

Motivations: Subgroups, Western Germany



By education | By age | By gender

Voters without higher education more Right-motivated in every election (band above 1/2 from 2021); AfD voters with higher education never predominantly right-wing. Young voters more Right-motivated in 2025 (van der Brug et al., 2026). Gender bands overlap (least variation in composition).

back

Robustness: Four Identification Checks



Municipalities $< 5{,}000$

Non-contender AfD districts

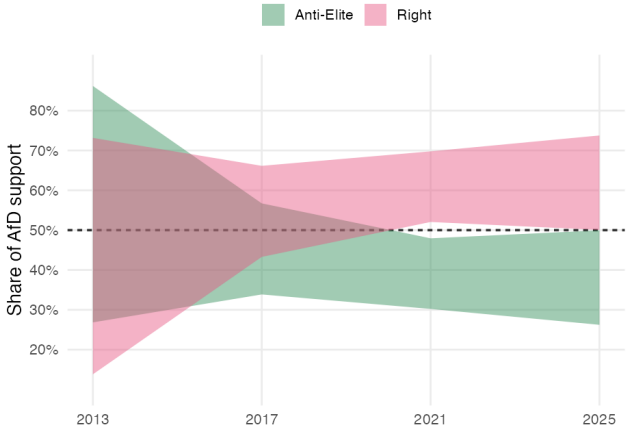
Conservative vs. moderate districts (masses may differ $\times 3$)

Border municipalities (equality within border segments)

Main qualitative results hold in all four; bounds wider in the last two, which impose weaker equality restrictions.

back

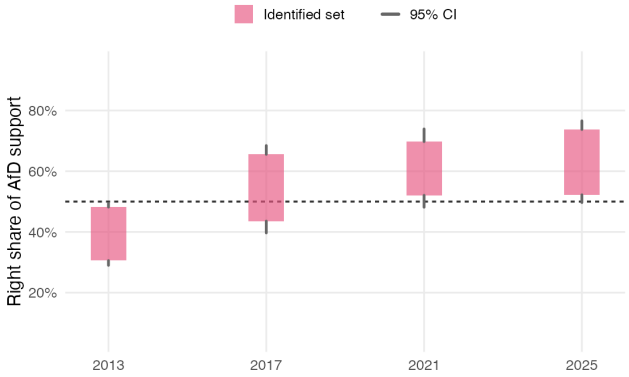
Motivations with Set-Identified Weights



Smallest lower / largest upper bound across all weights within 1% of the best fit. Results for 2017–2025 essentially unchanged.

back

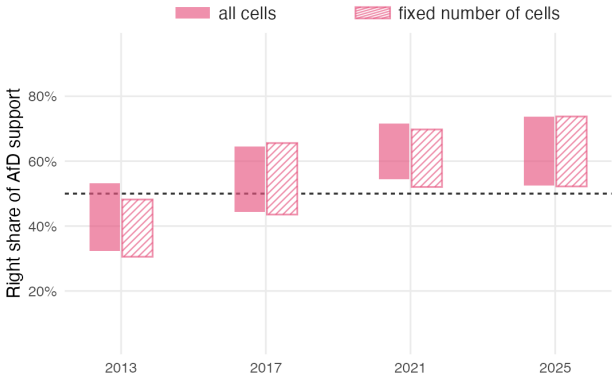
Motivations: Bootstrap Confidence Intervals



Bars: identified set for θ_{Right}; whiskers: 95% Imbens and Manski (2004) CIs, municipality bootstrap ($B = 100$).

back

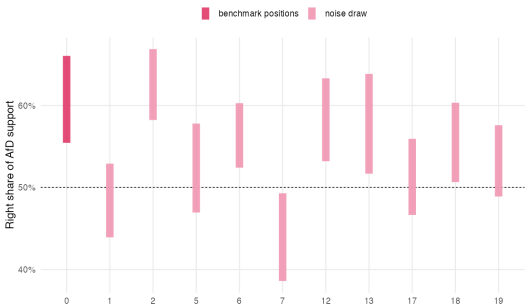
Full vs. Merged Partition



Solid: full MRP (sharp bounds); hatched: merged 1,000-cell mesh used in the main text. Close, with the full-cell bounds slightly narrower except 2013, at much higher computational cost.

back

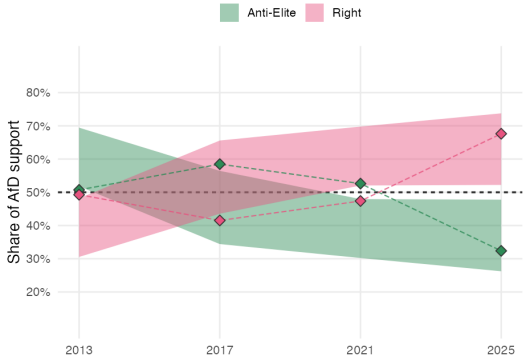
Measurement Error in Candidate Positions



West 2025: i.i.d. $N(0, 0.01^2)$ shocks to every candidate position, 10 replications. Shocks move the bounds without widening them and pull them towards the uninformative composition (attenuation-like). Positions measured with error enter as perturbed locations, not as an extra disturbance.

back

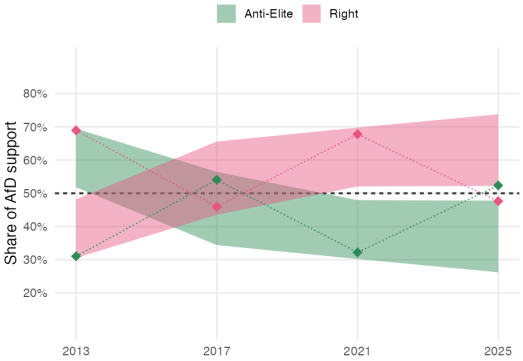
Stated Preferences: Motivations Exercise



Survey respondents assigned to the closer counterfactual party using the estimated region-year weights (weights cannot be recovered from survey data: small AfD samples; voters adjust stated positions towards their party). The survey reproduces the shift towards Right by 2025, but not the 2013–2017 change; attributes a larger role to Anti-Elite in 2017–2021, consistent with underreporting of right-wing views.

back

Comparison with Merlo and De Paula (2017) Point Estimates

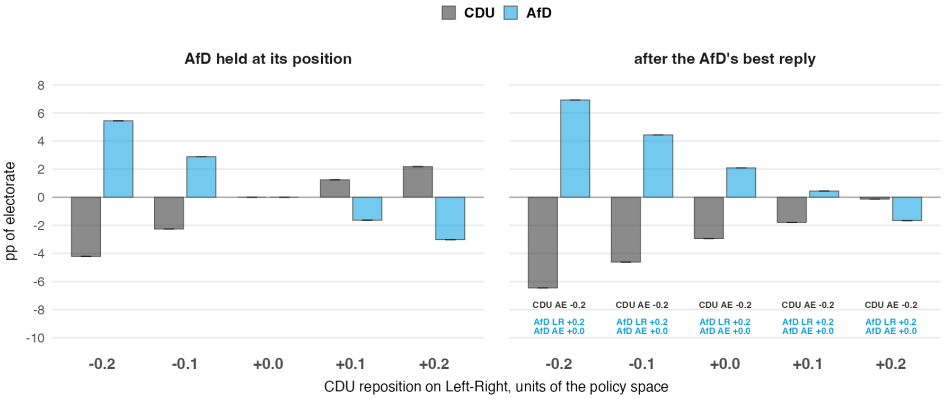


Sieve minimum-distance estimates (SNP density, order 2, anchored group means) vs. bounds

- Same data, groups, counterfactual masks; weight estimated jointly, agrees with the grid search
- Point estimates inside or close to the bounds in 2017–2025, not in 2013; no systematic trend – dominant motive switches between elections
- Depend on anchoring and polynomial order; the bounds do not
- In the game: AfD's best reply is always a step right (fitted density places large mass far right, pro-elite); both parties move right in equilibrium (next slide)

back

Repositioning Game under Merlo–de Paula Estimates



Left: CDU moves on Left–Right, AfD stands still. Right: CDU pairs each Left–Right move with its best Anti-Elite move, AfD replies on both dimensions; labels give the AfD's reply.

back

Shapley Decomposition: Details

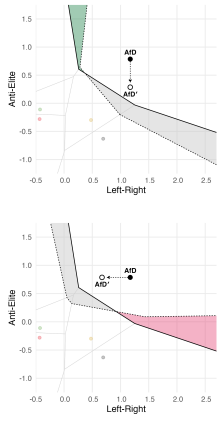
$\theta_{Right}(Pref_t, Comp_t, Pos_t)$: Right share given group-specific preferences $f(x \mid s, d)$, composition of AfD voters (Representative Electoral Statistics, 12 age–gender groups by state) and national median party positions Each Shapley value is the weighted average marginal contribution of a factor across all coalitions of the other two (weights $\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3}$); the three sum exactly to the change in θ_{Right}.

	2013→2017	2017→2021	2021→2025
West			
Composition	[−0.102, +0.141]	[−0.138, +0.119]	[−0.122, +0.106]
Positions	[−0.202, −0.026]	[−0.047, +0.000]	[+0.023, +0.117]
Preferences	[−0.287, +0.506]	[−0.432, +0.466]	[−0.345, +0.295]
East			
Composition	[−0.090, +0.048]	[−0.066, +0.042]	[−0.031, +0.045]
Positions	[−0.103, −0.018]	[+0.000, +0.018]	[−0.041, −0.005]
Preferences	[−0.384, +0.234]	[−0.233, +0.312]	[−0.302, +0.157]

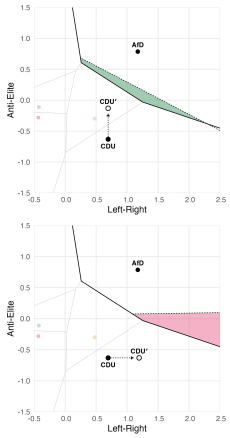
Each bound: min/max of the Shapley value over the identified set, one ideal-point distribution per year entering all counterfactual shares, AfD regions under both years' positions in the same LP.

back

Illustration: Counterfactual Shifts



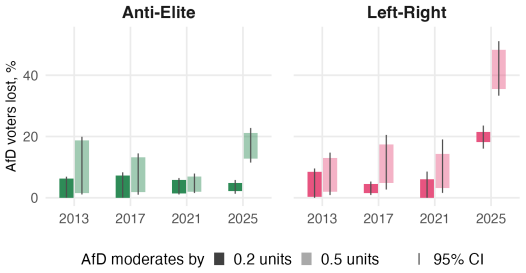
AfD moderates by 0.5 on Anti-Elite (top) / Left–Right (bottom); shaded: voters lost

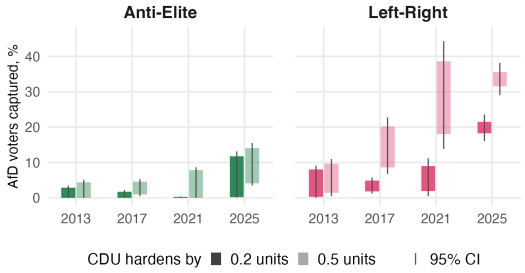


CDU hardens by 0.5 on Anti-Elite (top) / Left–Right (bottom); shaded: AfD voters captured

back

Repositioning: Bootstrap Confidence Intervals



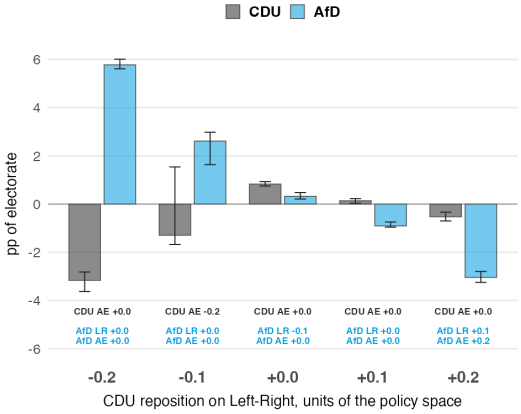


AfD moderates: share of AfD voters lost CDU hardens: share of AfD voters captured

Bars: identified sets; lines: 95% Imbens and Manski (2004) CIs, municipality bootstrap ($B = 50$), truncated at zero.

back

Equilibrium in Both Dimensions

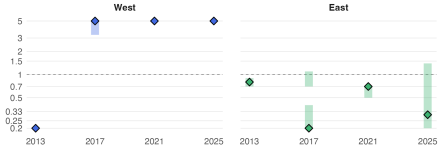


Net changes after the AfD's best reply when both parties move on both dimensions (CDU paired with its best Anti-Elite move). CDU's $+0.2$ leaves the CDU at -0.5 and the AfD at -3.0 pp at the median. Labels: CDU's Anti-Elite move and the AfD's modal reply as (LR, AE).

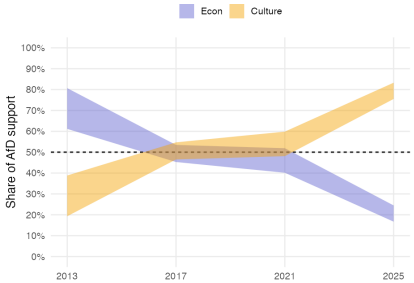
back

Left–Right as Economics × Culture

Re-estimate the model with **Economics** and **Culture** (immigration coded as cultural, self-placement dropped) as the two dimensions; omitting Anti-Elite is innocuous since it is weakly correlated with both



Weight on Culture ($w_{econ} = 1$)



Economically vs. culturally motivated AfD support

Gradual realignment from economic motives (dominant in 2013, "professors' party") to cultural ones (dominant in 2025); West attaches much larger weight to culture from 2017 back

Firewall against Cooperation with the Far Right

Apply the estimated 2025 preferences to the two numerically feasible coalitions, CDU–SPD and CDU–AfD (policy = midpoint of members' positions):

- **Utilitarian welfare** (average weighted distance from ideal points): national bounds straddle zero, $[-0.18,\ 0.03]$ – neither coalition dominates; strictly favor CDU–AfD in the East
- **Majoritarian choice:** for every distribution in the identified set a clear majority is closer to the CDU–AfD policy (61.9–72.8% nationally; $\geq$ 69.9% East, $\geq$ 58.5% West)
- The gap reflects intense opposition to CDU–AfD among voters on the left; whether the firewall leaves voters underrepresented depends on the aggregation rule, sharpest in the East

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Probabilistic Voting and Omitted Dimensions

The model as a probabilistic voting model. With X_i the type's average ideal point and ϵ_i a voter-level deviation,

$$U_j^{RPM} = -(X + \epsilon - Y_j)^2 = -(X - Y_j)^2 + \underbrace{2\epsilon(Y_j - X) - \epsilon^2}_{\nu_j}$$

a random utility model whose error distribution is *induced* by the ideal-point shock (correlated across candidates) rather than assumed. A taste component orthogonal to (X_i, Y_j) does not bias ideal points, but may inflate their dispersion.

Omitted dimension with weight w^o and positions $Y_j^o = \gamma Y_j$: the recovered ideal point is $\tilde{X} = \frac{wX + w^o\gamma X^o}{w + w^o\gamma^2}$ and the recovered weight $\tilde{w} = w + w^o\gamma^2$. Distortions vanish if $\gamma \to 0$ (unrelated positions) or $w^o \to 0$ (unimportant). Left–Right and Anti-Elite are essentially uncorrelated and jointly explain 72% of the variance of party positions across eight CHES dimensions.

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